PROGRAMMING ASSIGNMENT

Tithira Withanaarachchi(Batch 04)-0119

Lecturer –
Mahesha Thejani

Task 1[12]

.NET Framework is a platform that provides facilities to create different applications. There are many components which supports the activities of the platform. Among them there are 2 main components called,

1. Common Language Runtime. (CLR)
2. .NET Framework Class Library. (FCL)

1. **Common Language Runtime. (CLR)**

   .NET Framework allows to run programs through an environment called Common Language Runtime. It allows to run all types of .NET programs. Code which executes through this is called a Managed Code. It provides both memory management and threat management which will make it more convenient for the user to handle with the application development using .NET framework. Language compilers convert the program to Microsoft Intermediate Language (MSIL) intern CLR converts this to Native code.

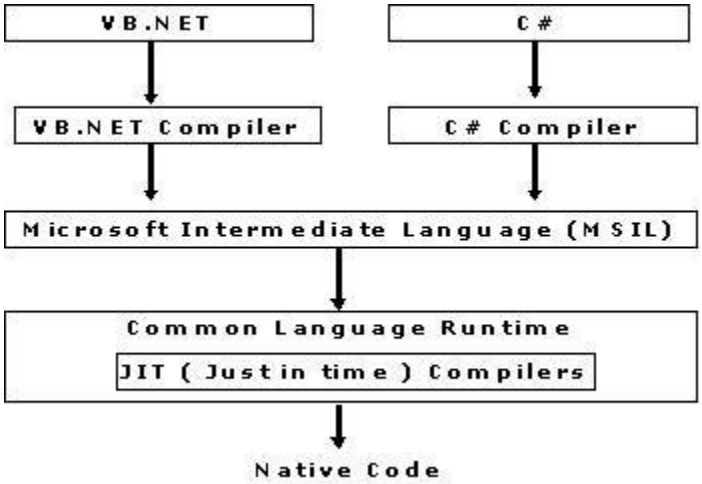


Figure 1 Conversion of the program to the Native code

2. .NET Framework Class Library. (FCL)

This is also known as the base class library which is common for all the types of applications.

Below are some applications that use .NET class library,

1. Windows Application.
2. Console Application.
3. Web Application.
4. XML Web Services.
5. Windows Services.

In short, developers just need to establish the Bass class library in their language code and use its predefined methods to implement from simple functions to very complex functions as well.

There are also more other supportive components in the .NET Framework other that the above main components such as,

- Common Type System,
- Common Language Specification,

Common Type System –

It describes the datatypes that can be used in various .NET languages in common. It is the component which make sures that different .NET languages can interact with each other.

Common Type System Supports 2 standard Categories of types,

- Value types,
- Reference Types.

Common Language Specification –

It explains minimal or complete set of features that can be made by CLR. It make sure that products of compilers work properly

It describes the minimal and complete set of features to produce code that can be hosted by CLR. It ensures that products of compilers will work properly in .NET environment.

Sample Rules:

1. Representation of text strings

2. Internal representation of enumerations

3. Definition of static members and this is a subset of the CTS which all .NET languages are expected to support.[12]

Task 2

Flow Chart for Login Function.

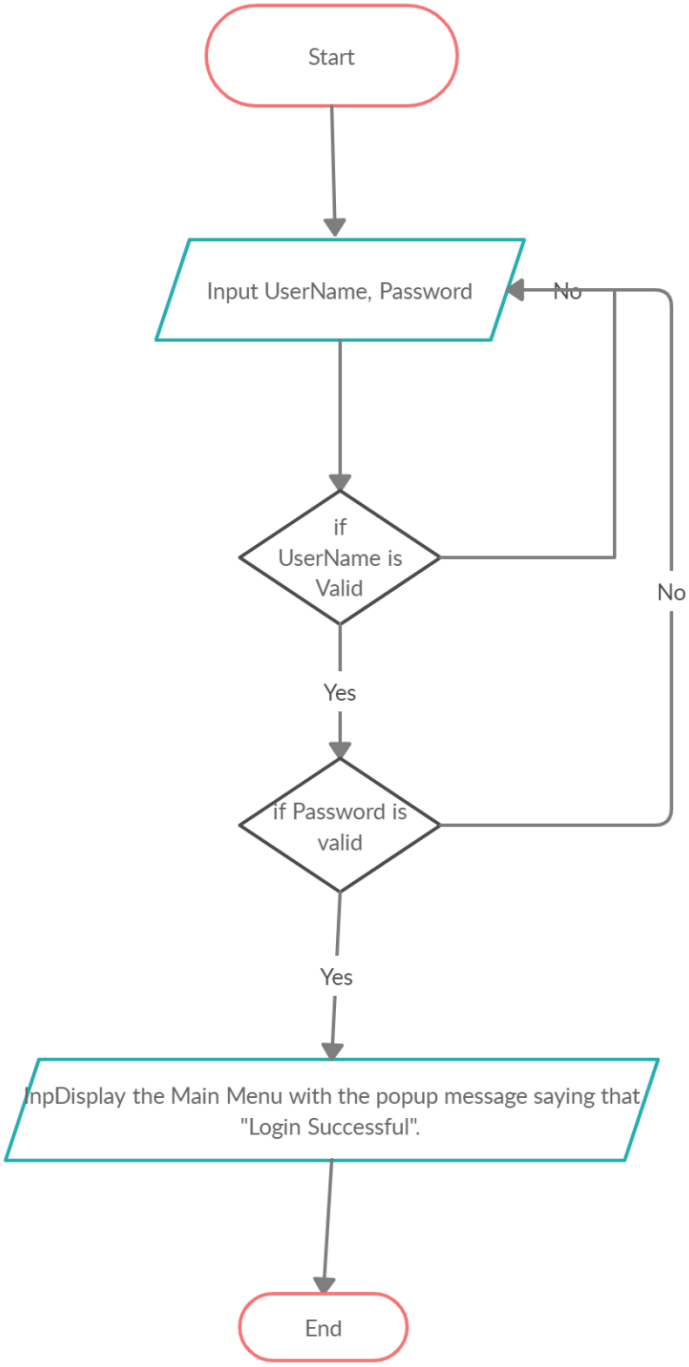


Figure 2 Flow Chart - Login

Flow Chart for Insert Function

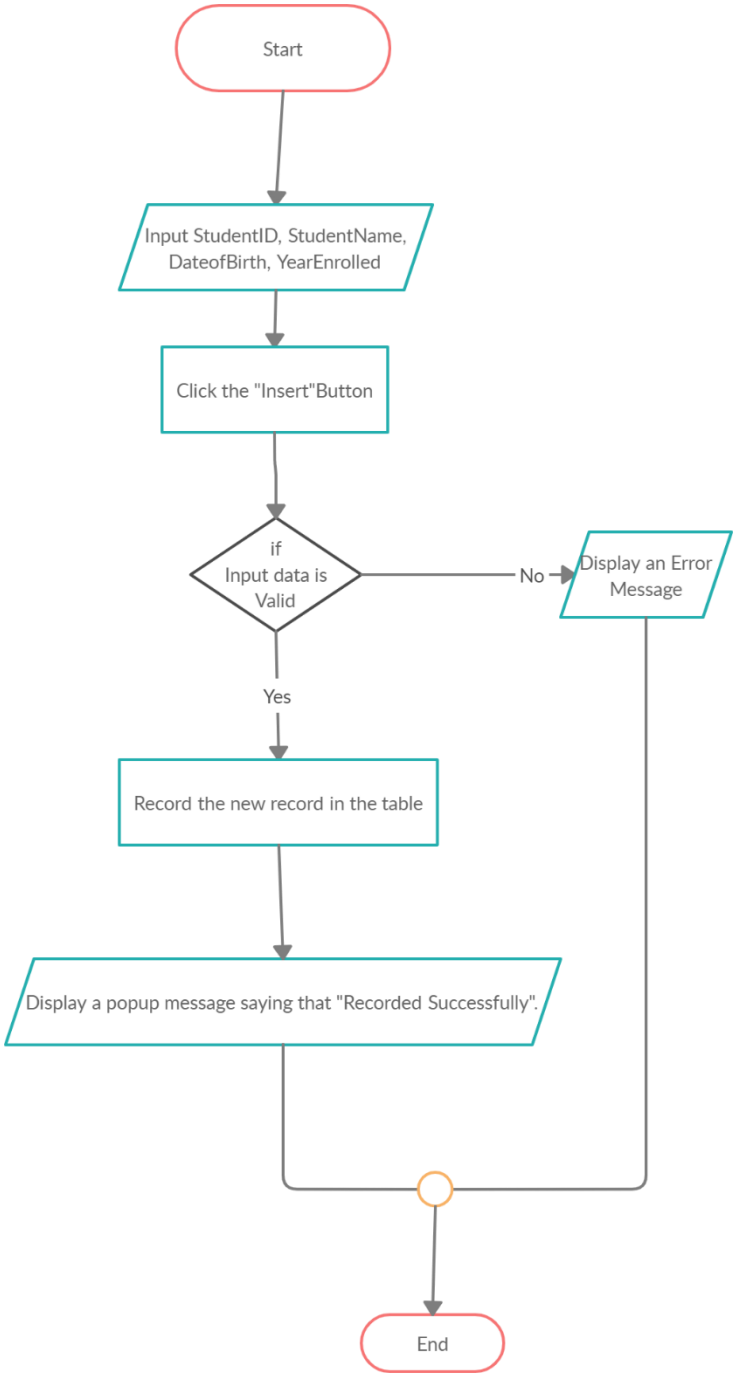


Figure 3 Flow Chart - Insert

Flow Chart for Delete Function.

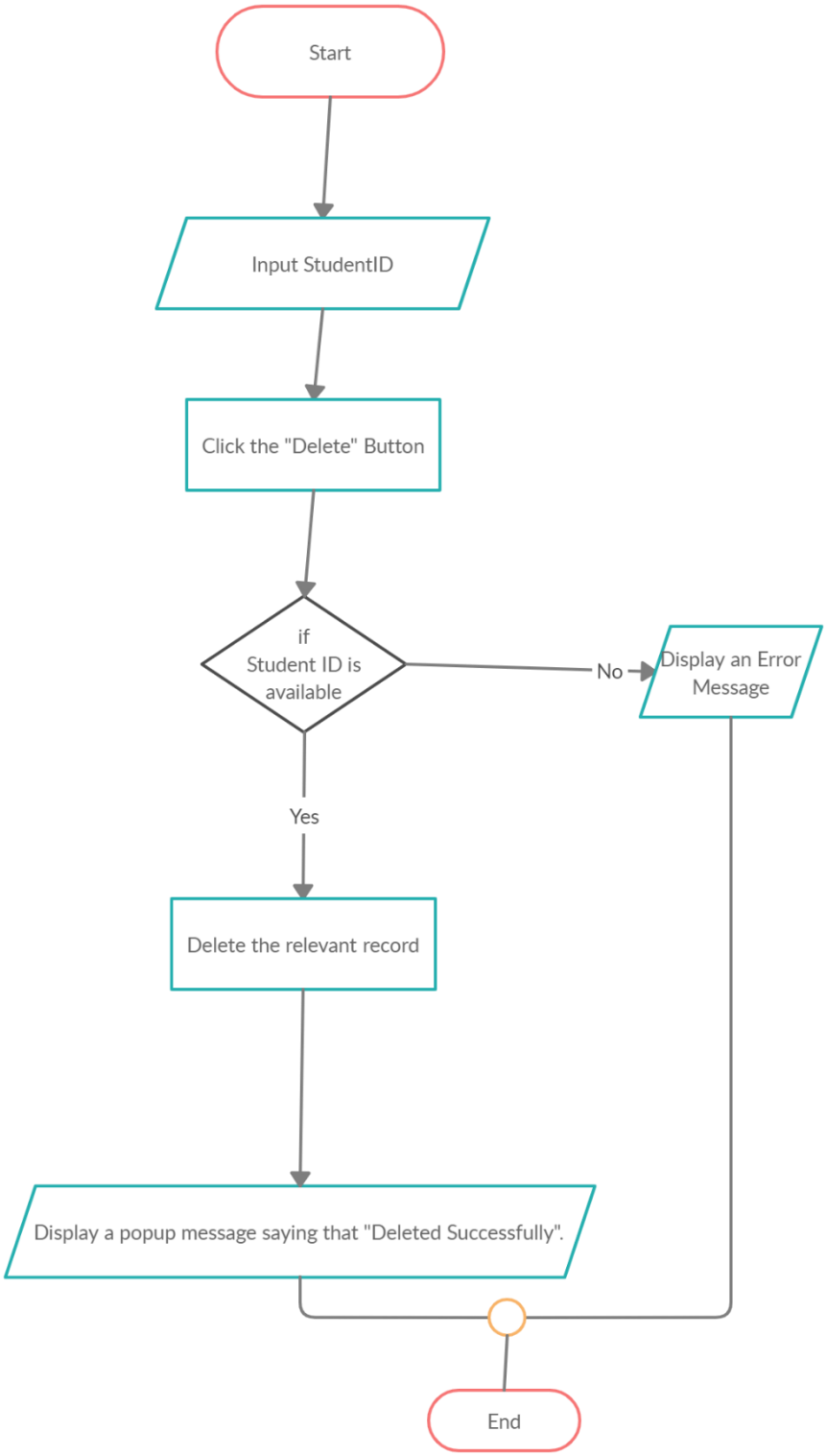


Figure 4 Flow Chart - Delete

Flow Chart for Search and Update.

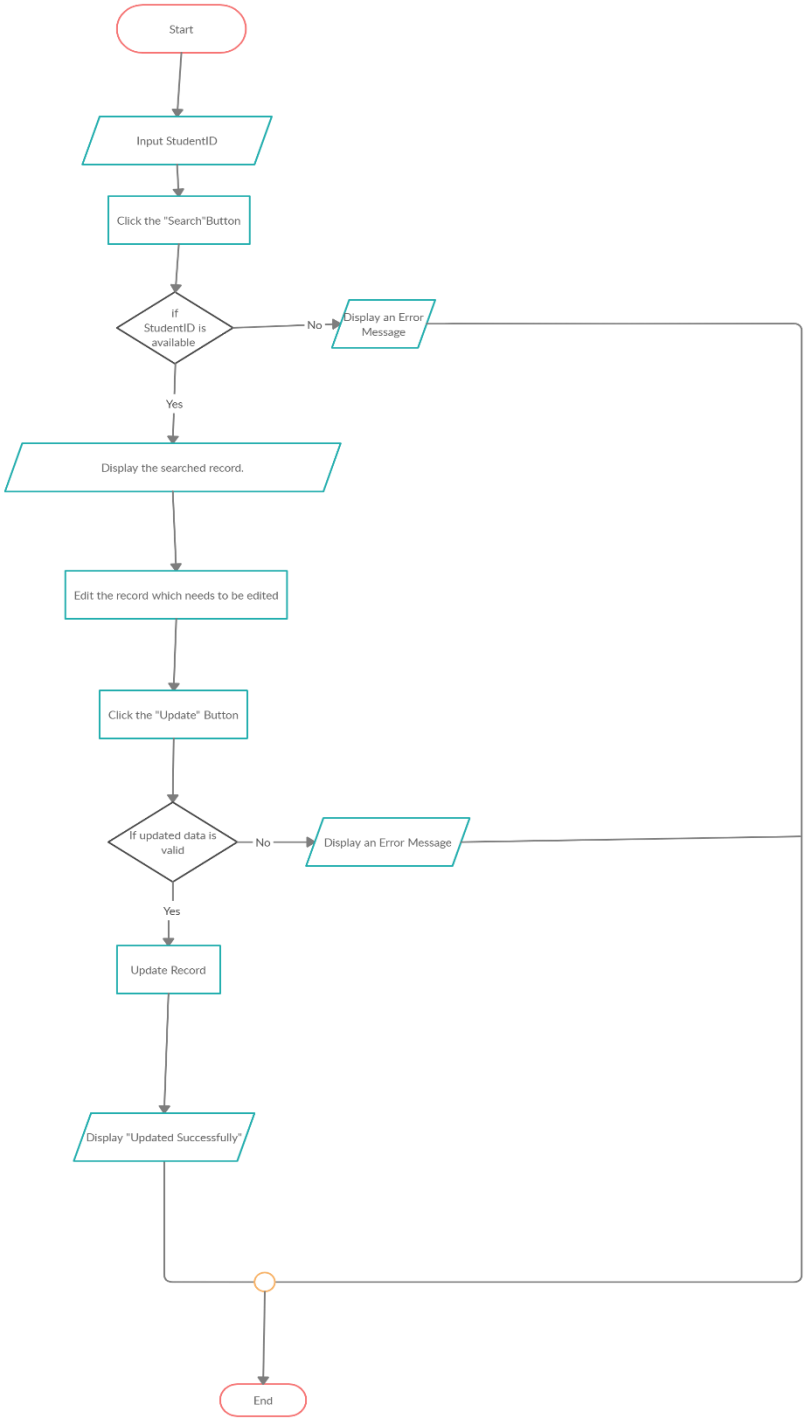


Figure 5 Flow Chart Search and Update

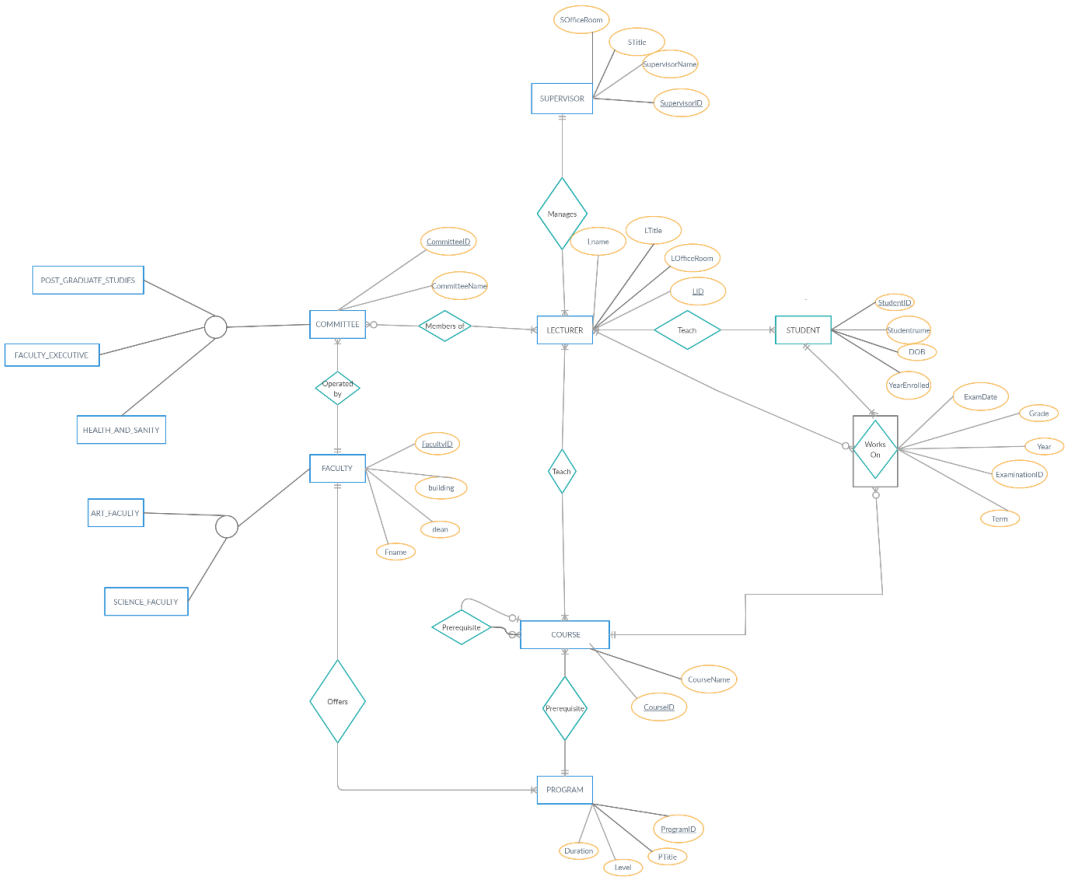


Figure 6 ER Diagram for Information Management System of Stafford University

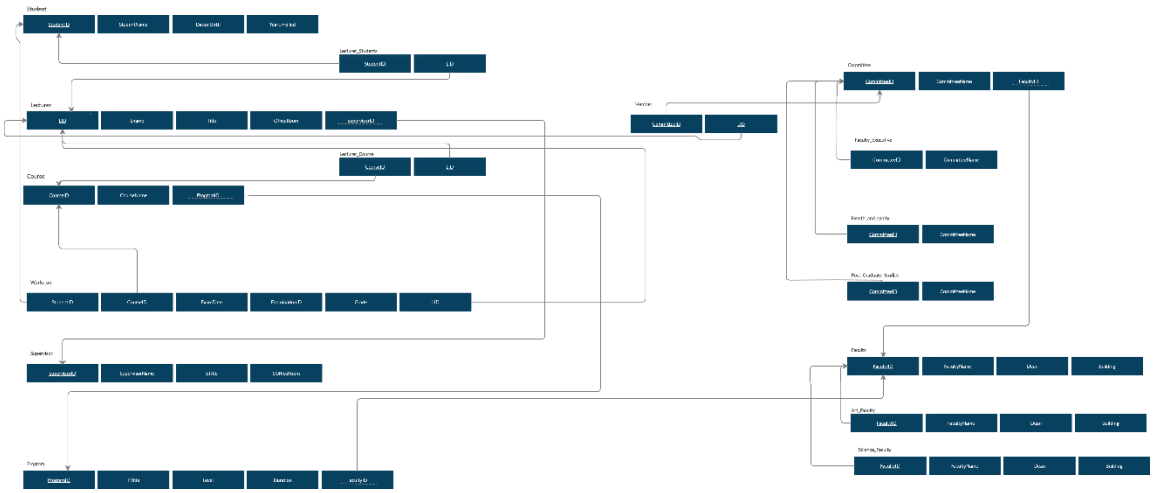


Figure 7 Relational Schema for the Information Management System of Stafford University

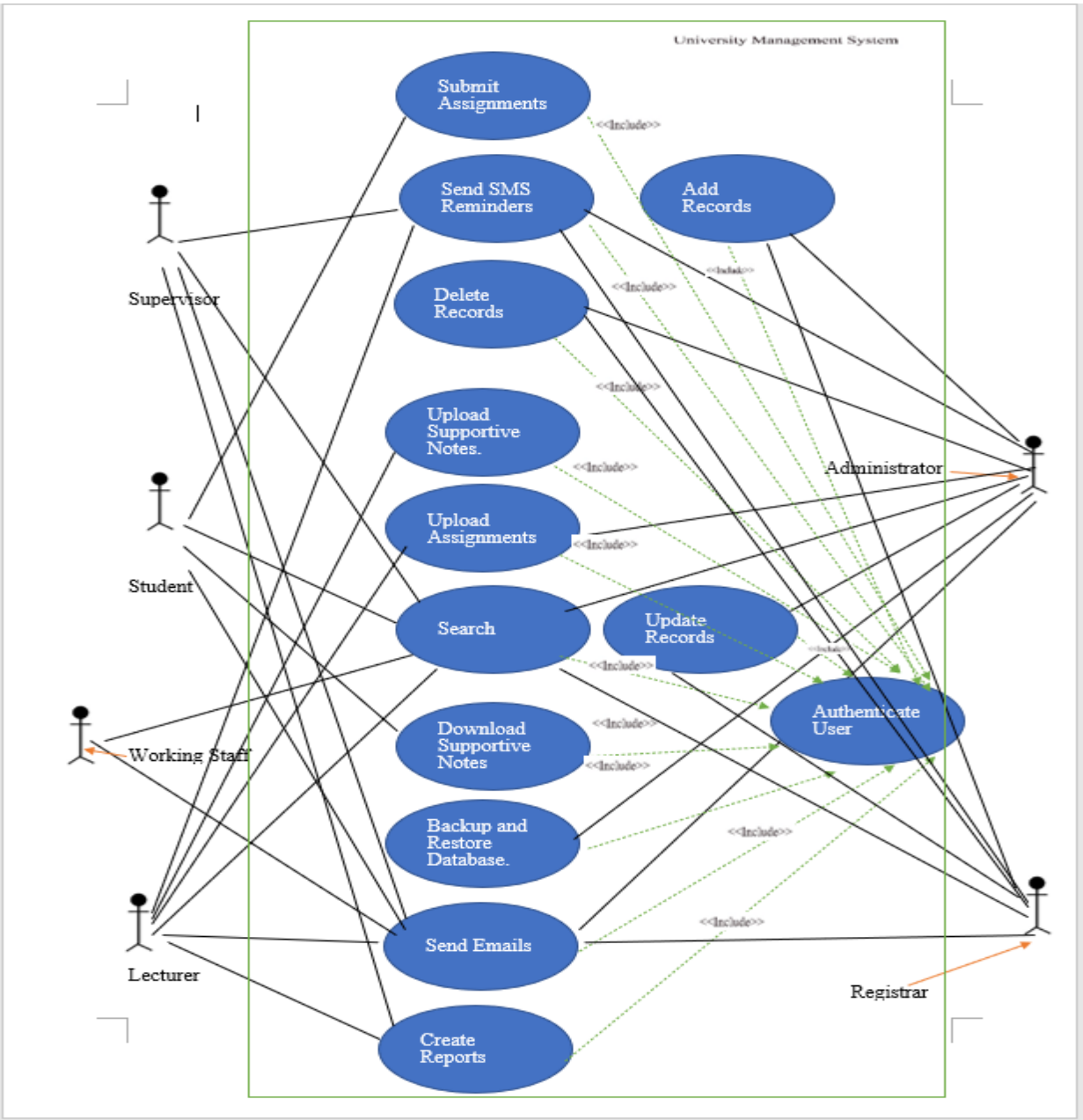


Figure 8 Use Case Diagram for the Information Management System of Stafford University

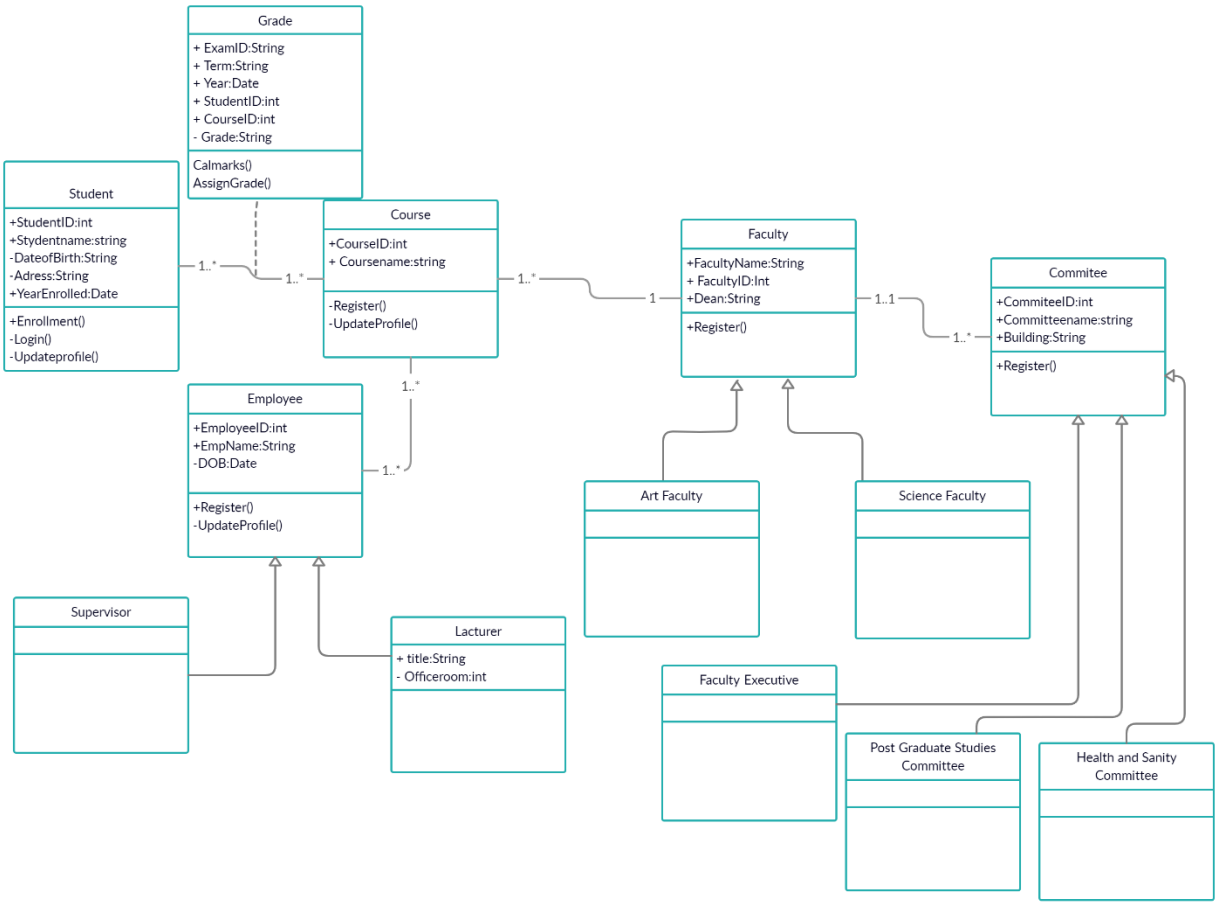


Figure 9 Class Diagram for the Information Management System of Stafford University

Task 3

3.1)

Example for Control structures (from the implemented code):

```
1 reference
private void button1_Click(object sender, EventArgs e)//Button Click for Login
{
    try
    {
        SqlDataAdapter sda = new SqlDataAdapter("Select count(*) from login where username = '" + textBox1.Text + "' and password = '" + textBox2.Text + "'", m_con);/*Sql Query
        to count Number of records available according to the text input(it can be 1 or 0 only)*/
        DataTable dt = new DataTable();//Creating a virtual table
        sda.Fill(dt);//filling selected value from the sql query to the virtual table dt
        if (dt.Rows[0][0].ToString() == "1")//Control Structure: checking whether there is a record tallying with the input data.
        {
            this.Hide();

            MainMenu ob = new MainMenu();//Creating new object from main menu
            ob.Show();//Display the main menu

        }
        else
        {
            MessageBox.Show("Invalid Access", "Stafford University Management System");//Display error message
        }
    }
    catch
    {
        m_con.Close();
    }
    /*Reference - https://www.c-sharpcorner.com/UploadFile/9582c9/login-form-with-sql-in-C-Sharp/ - (Online website)*/
```

Figure 10 Example for Implemented control structures - Extracted from the implemented code (Login) - Reference - [1]

The above code is for the Login button of the login interface which allows the admin to login to the Information Management System of Stafford University after validating the entered Password and Username.

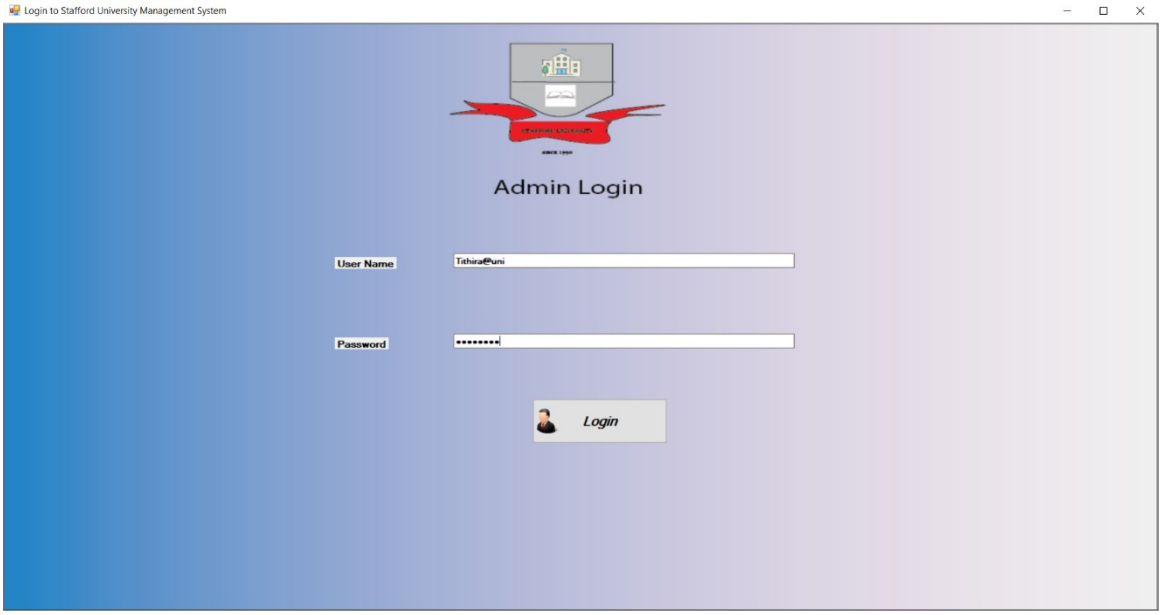
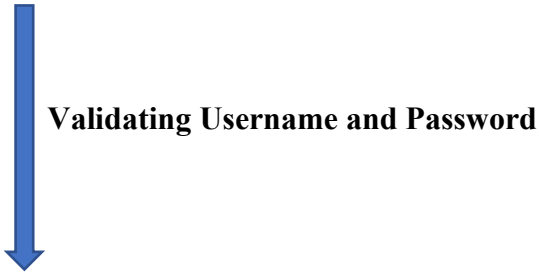


Figure 11 Login Interface - (Extracted from the implemented system) [1],[3], [5]



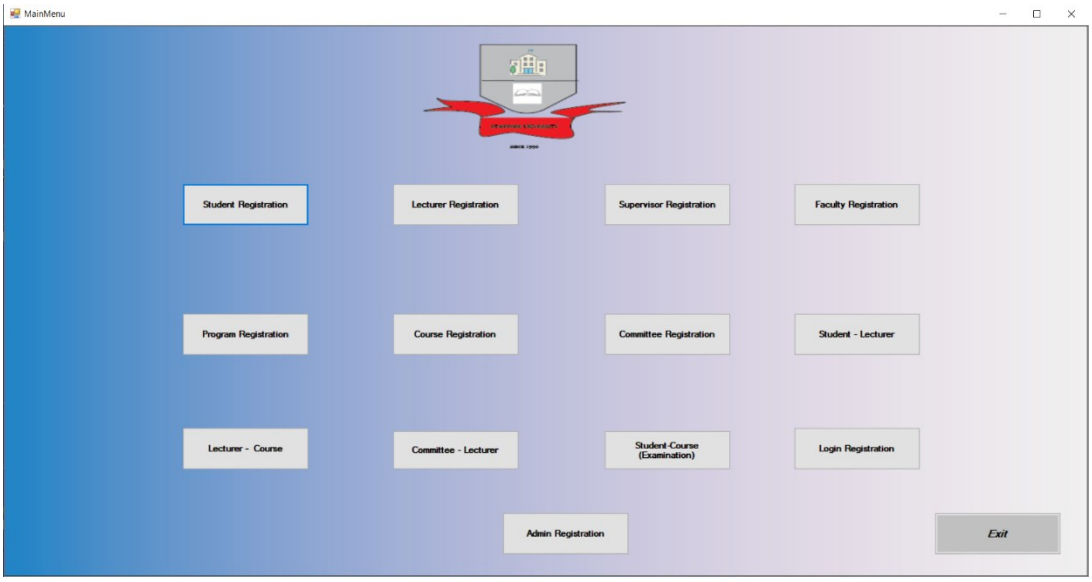
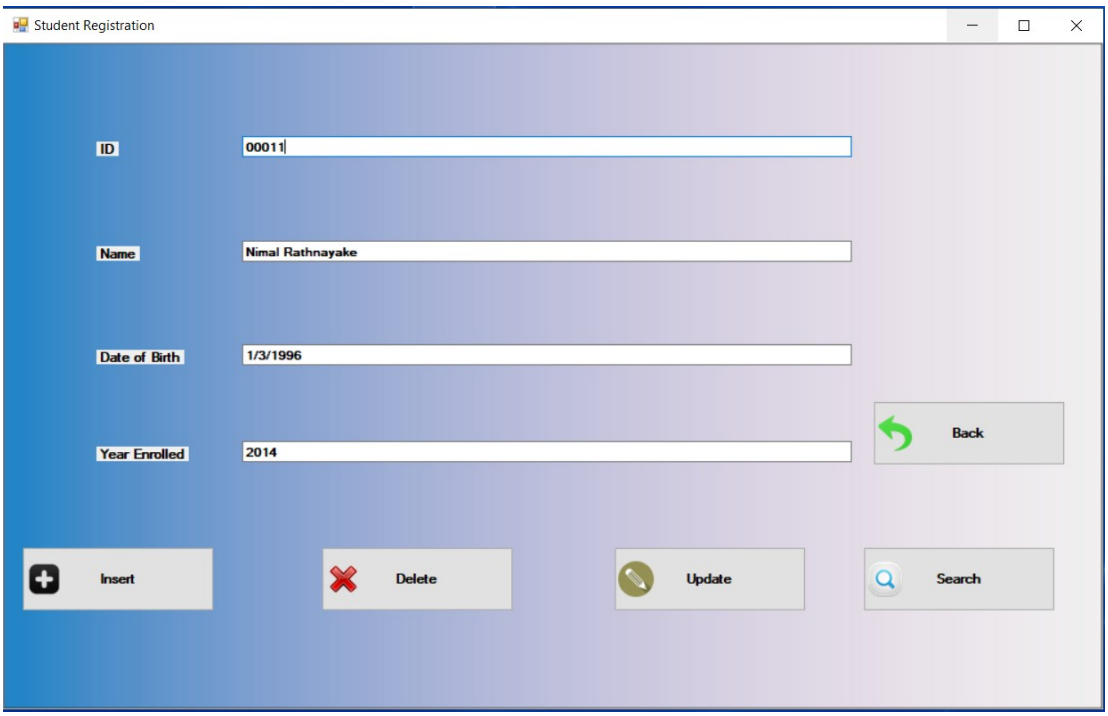


Figure 12 Main Menu - (Extracted from the Implemented system)

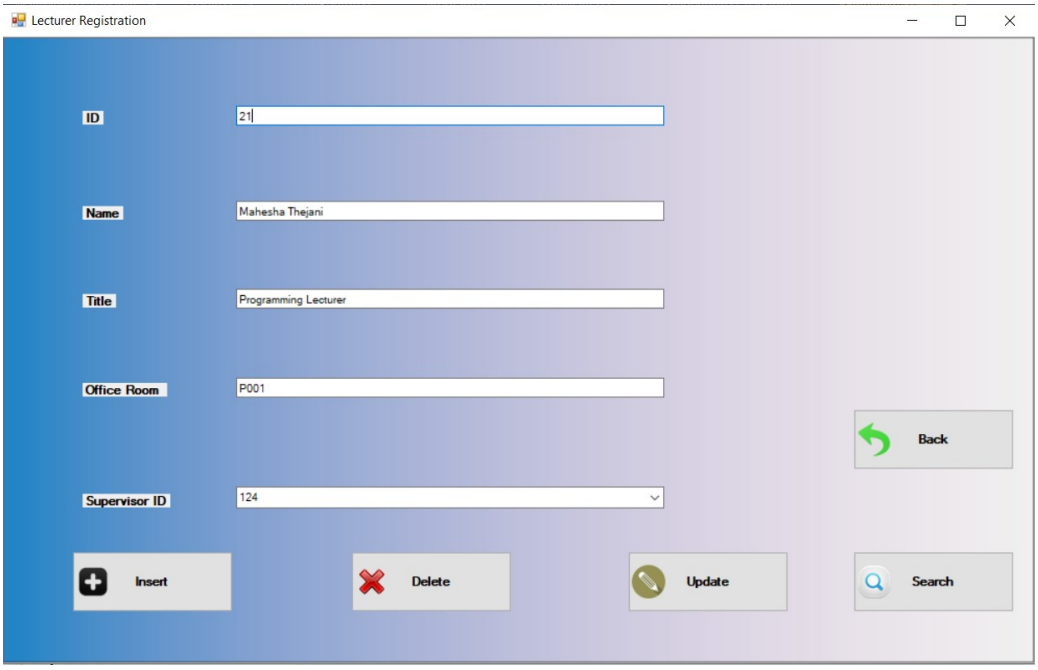
When user enter a valid username and a password in the text boxes given for each in the login interface (Figure 2) and click the login button, user will be directed to the Main Menu Interface (Figure 3).

Main Menu includes 13 buttons which directs the user for another interface which allows the user to Insert, Delete, Update and search data according to their table names in the database.

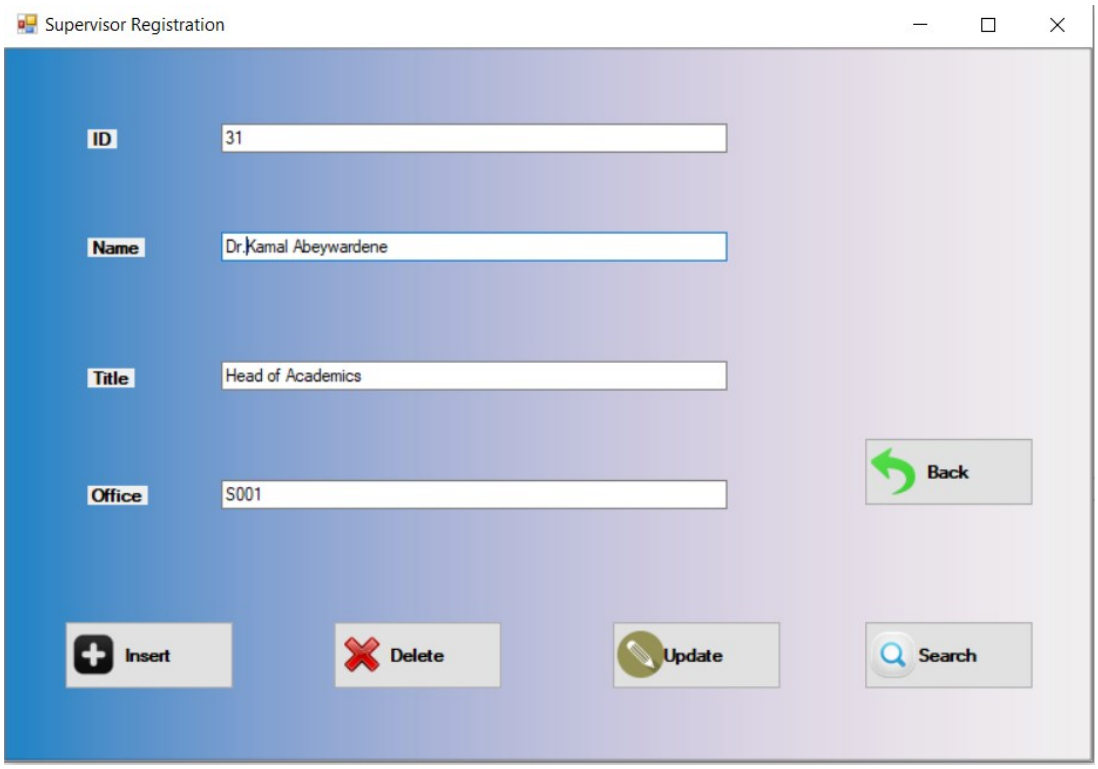
1. Student Registration Interface.



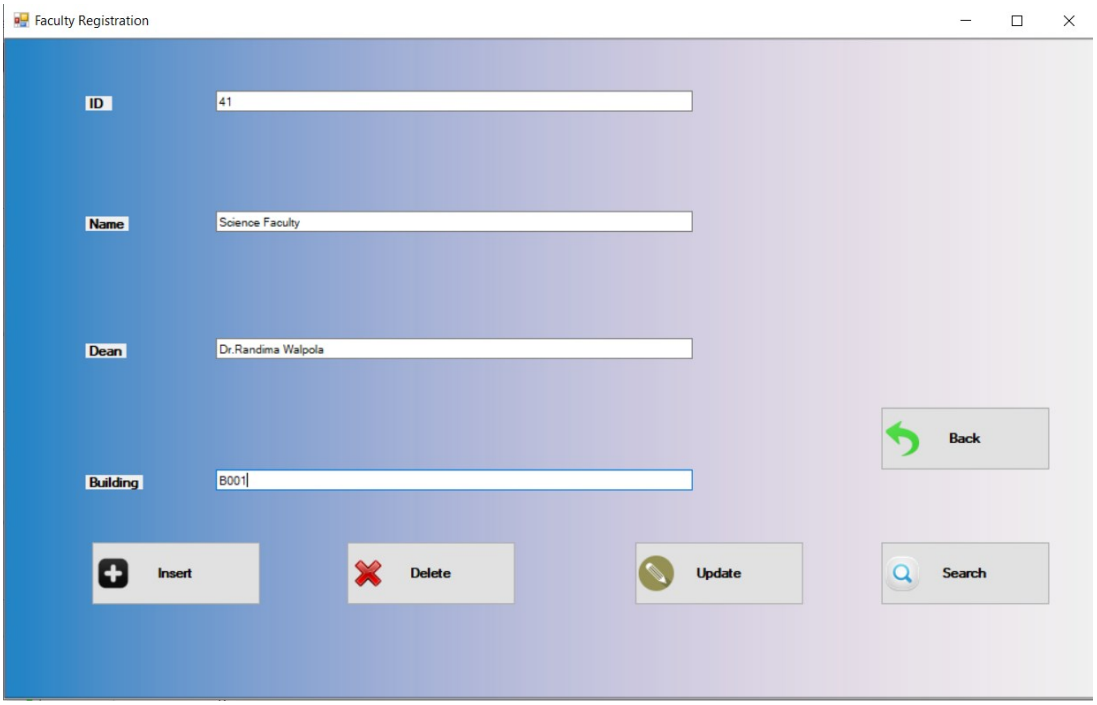
2. Lecturer Registration Interface



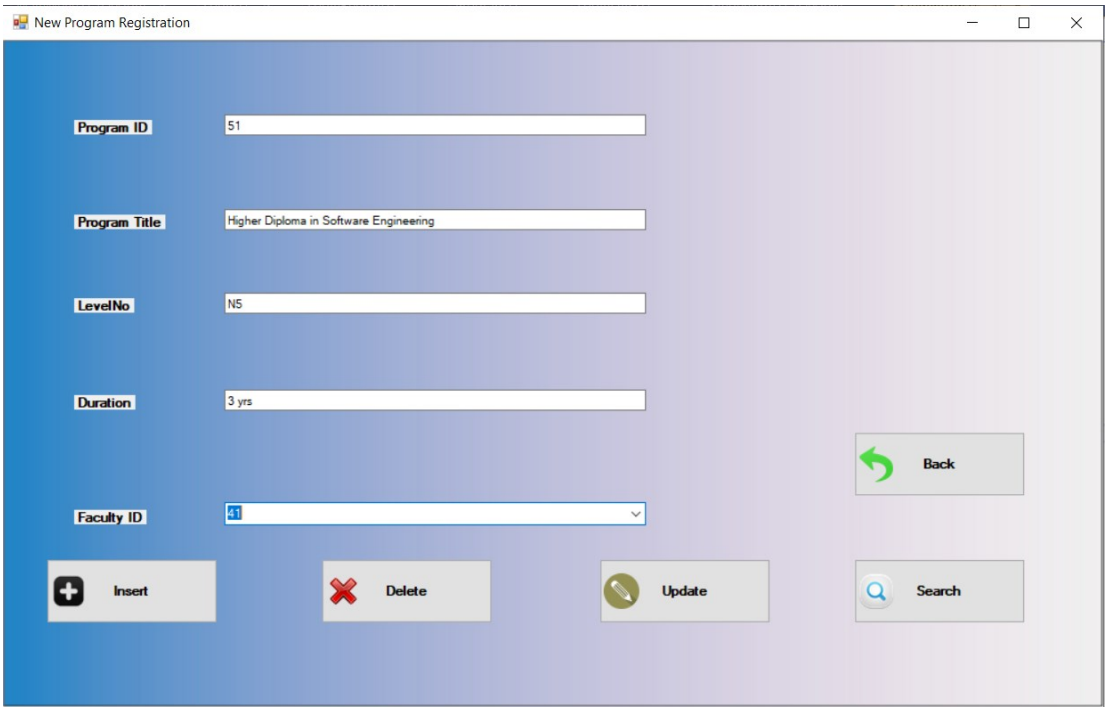
3. Supervisor Registration Interface



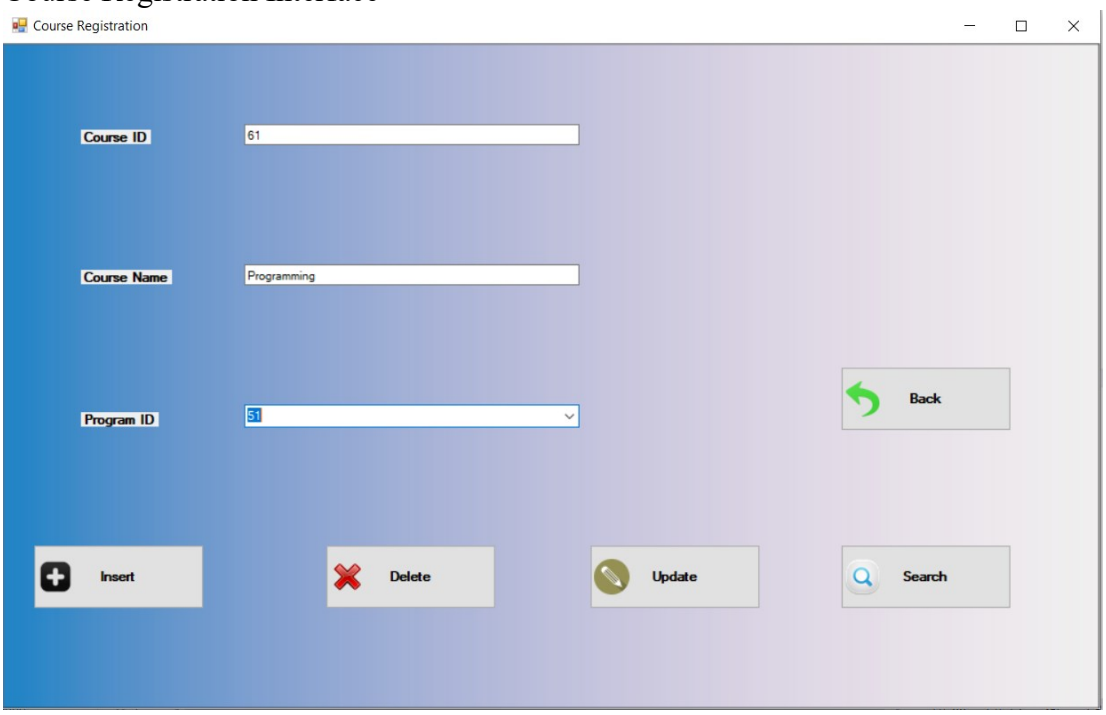
4. Faculty Registration Interface



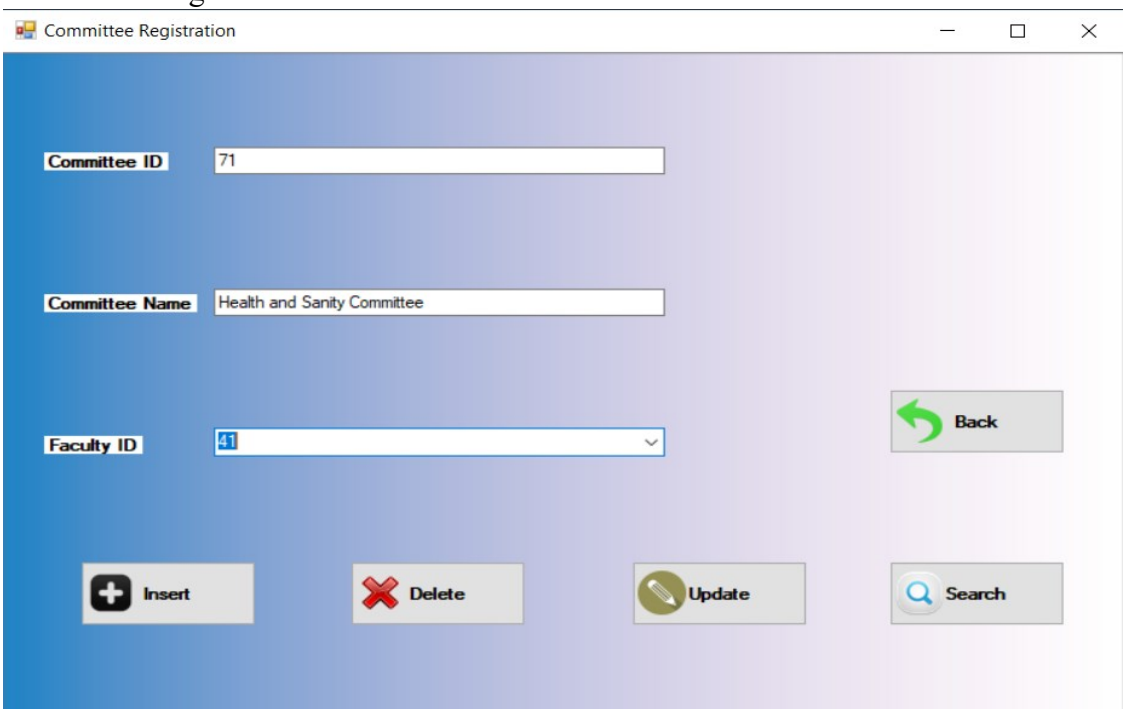
5. Program Interface –



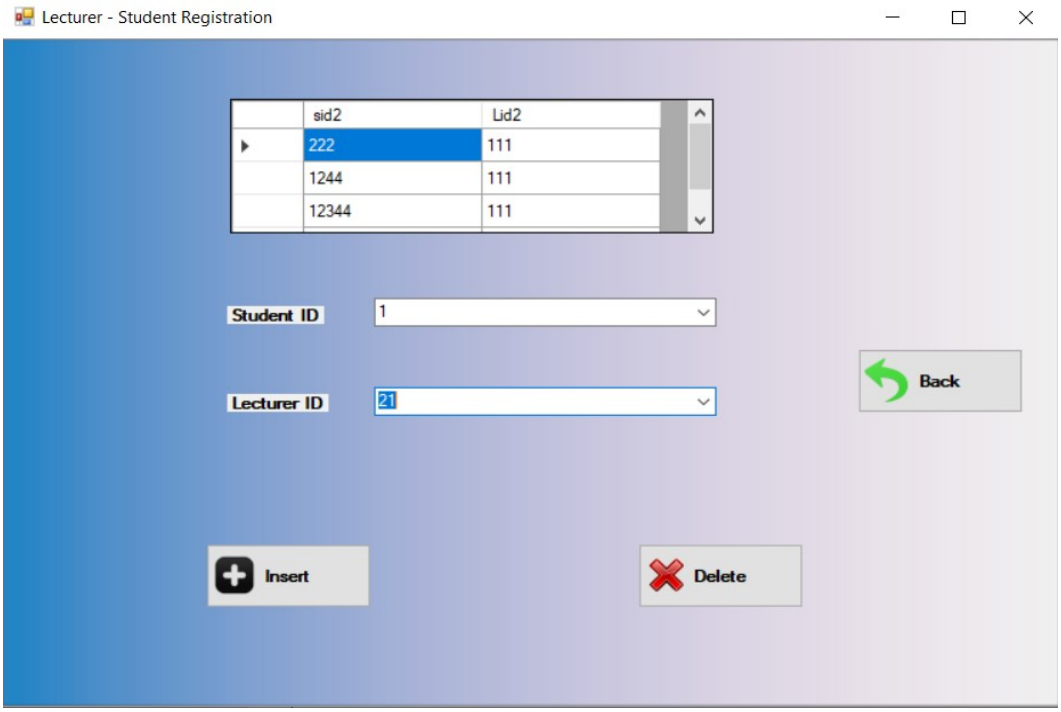
6. Course Registration Interface –



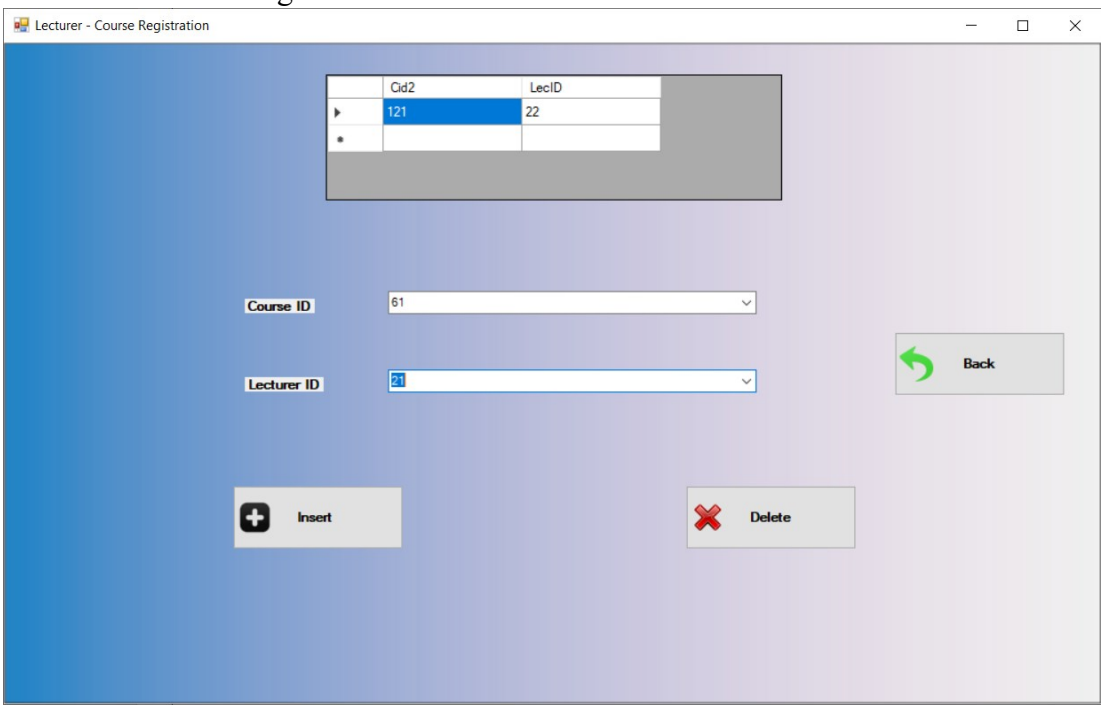
7. Committee Registration Interface –



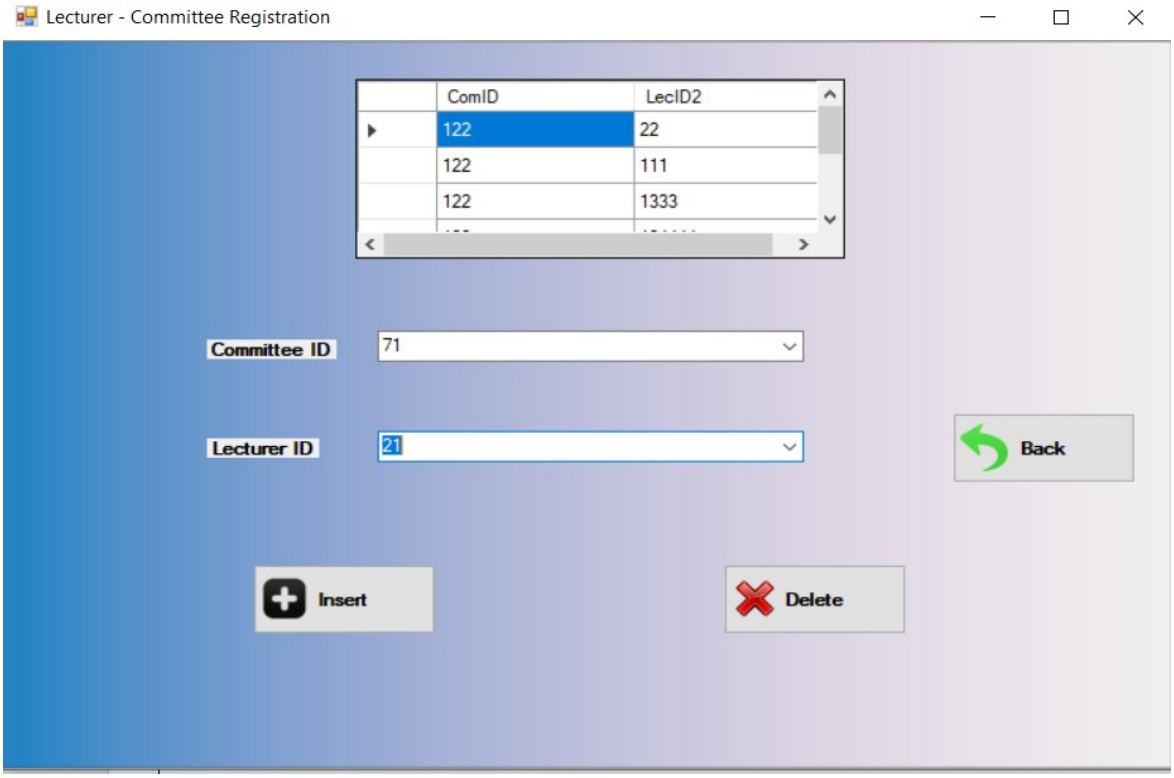
8. Student – Lecturer Registration Interface –



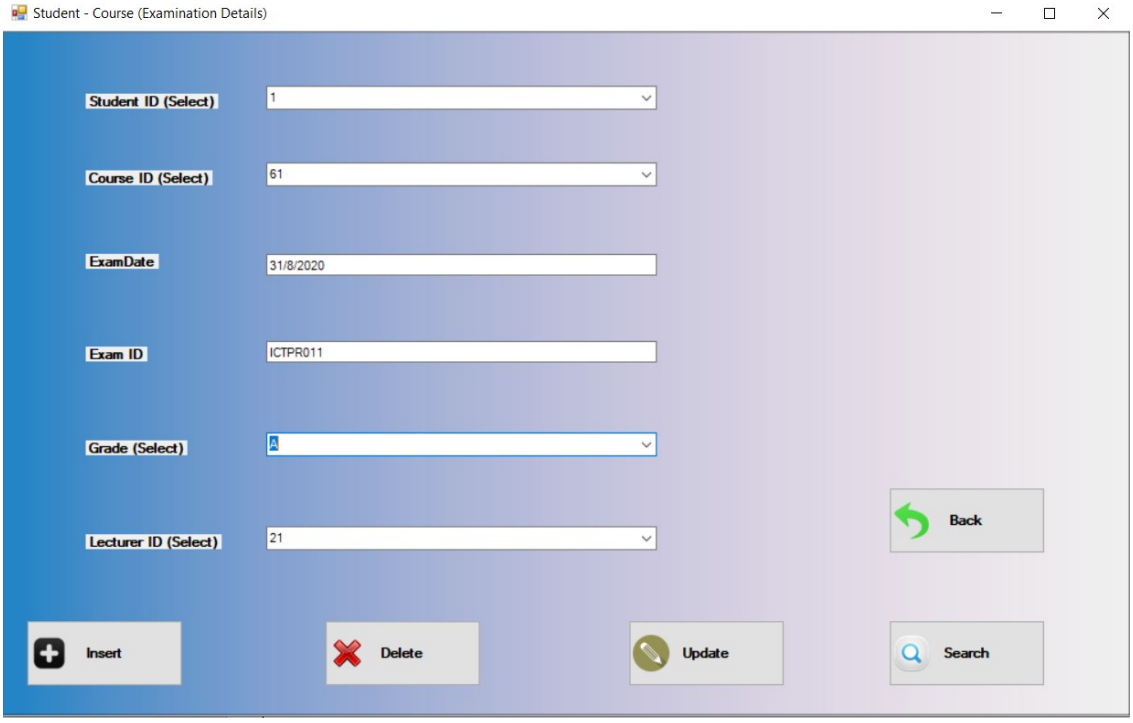
9. Lecturer – Course Registration Interface



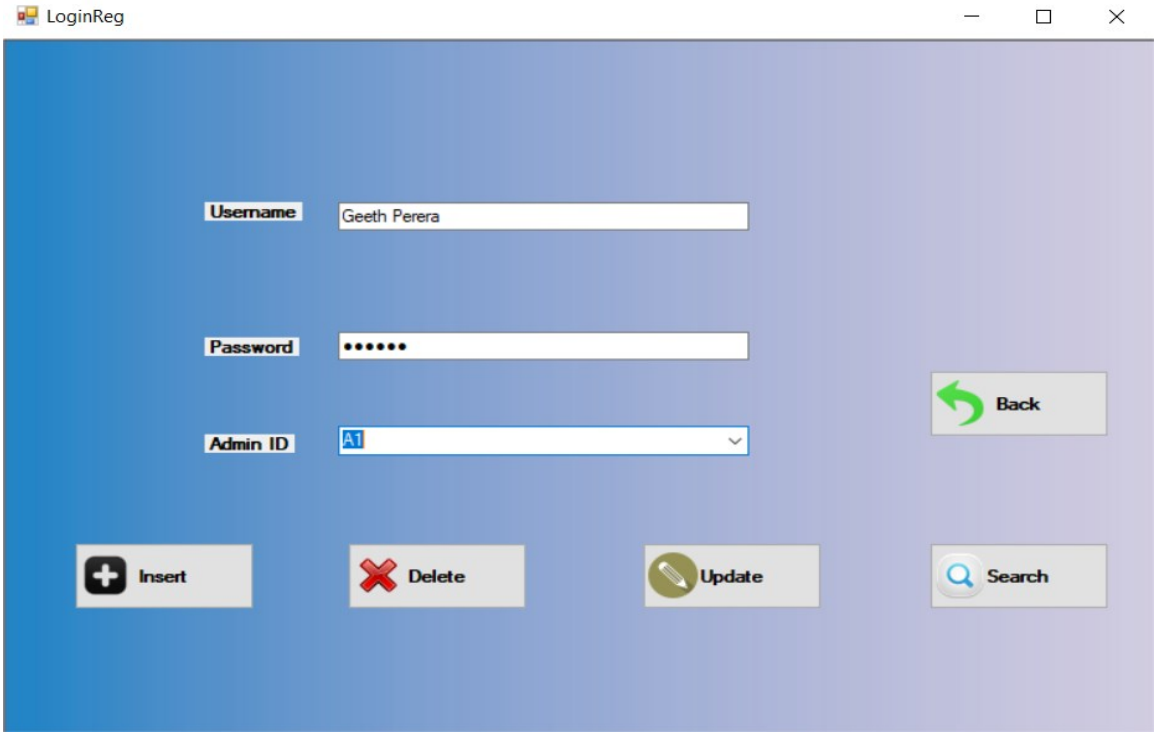
10. Lecturer - Committee Registration Interface –



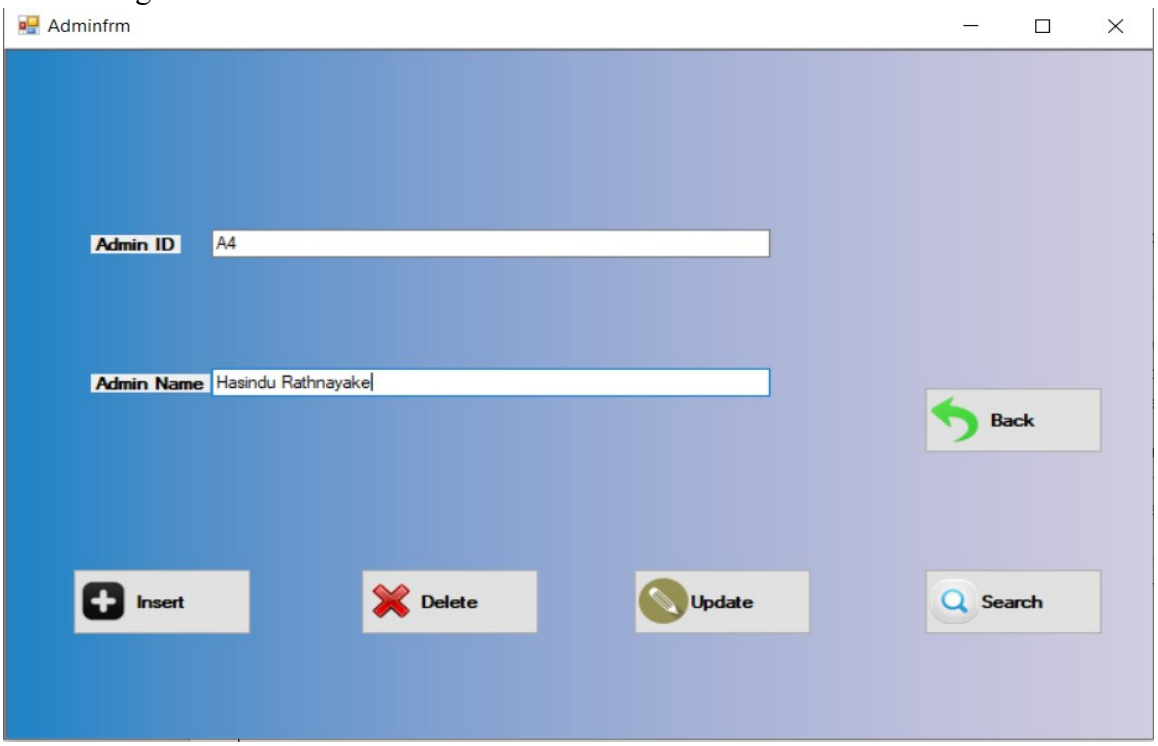
11. Student - Course Registration Interface –



12. Login Registration Interface –



13. Admin Registration Interface –



```
namespace Stafford_Uni_Management
{
    2 references
    class StudentDAO
    {
        Student std1;//object from the student class
        static string con = "Data Source=DESKTOP-2ORO7LR\\SQLEXPRESS;Initial Catalog=Tithi;User ID=Tithi;Password=tithira@sql";//sql connection string
        SqlConnection m_con = new SqlConnection(con);//sql connection
```

- Given above is the coding part which is used to make a connection string to the database from the c sharp file.

```
public void insertnewstudent(Student sobj)//method to insert new records to the student table in the database
    {
        try
        {
            String sql = "insertintoStudent(StudentID,StudentName,DateofBirth,YearEnrolled)values(" + sobj.getid() + ",'" + sobj.getname() + "', '" + sobj.getdob() + "', '" + sobj.getyenrolled() + "')";//sql insert query for the student table in the database

            SqlCommand cmd = new SqlCommand(sql, m_con);//sql command to connect to Database

            m_con.Open();

            cmd.ExecuteNonQuery();

        }
        finally
        {
            m_con.Close();
        }
    }
```

- Above Code is the Public Void Method written in the studentDAO class and used to insert new records to the Student Table in the Database.
- This method is written in all other DAO Classes related to entities of the Information Management System which includes, Lecturer Registration, Supervisor Registration, Faculty Registration, Program Registration, Course Registration, Committee Registration, Student – Lecturer Registration, Lecturer - Course Registration, Lecturer - Committee Registration, Student - Course (Examination Details), Login Registration, Admin Registration.

```
public void deletestudentbyid(int id)//method used to delete records from student table
    {
      try
      {
        string sql = "delete from student where studentid = " + id + "";//sql
        query used to delete records from student table

        SqlCommand cmd = new SqlCommand(sql, m_con);//sql command to make the
        connection with database

        m_con.Open();

        cmd.ExecuteNonQuery();
      }
      finally
      {
        m_con.Close();
      }


    }
```

- Given above is the public void method written in the studentDAO class and used to delete records from the database.
- This method is also written in all other DAO classes related to entities of the Information Management System which includes, Lecturer Registration, Supervisor Registration, Faculty Registration, Program Registration, Course Registration, Committee Registration, Student – Lecturer Registration, Lecturer - Course Registration, Lecturer - Committee Registration, Student - Course (Examination Details), Login Registration, Admin Registration.

```
public void updatestudent(Student sobj)//method used to update student records in the student
                                        table of the database
    {
      try
      {
        string sql = "update student set studentname = ('" + sobj.getname() + "'), dateofbirth =
                    ('" + sobj.getdob() + "'),yearenrolled = ('" + sobj.getyenrolled() + "')where
                    studentid = ('"+sobj.getid()+"')";//sql query part used to update student
                    records in the student table of database

        SqlCommand cmd = new SqlCommand(sql, m_con);//sql command used to connect to
                                                        the database

        m_con.Open();

        cmd.ExecuteNonQuery();
      }
      finally
      {
        m_con.Close();
      }
    }
```

- The coding part given above is written in the StudentDAO class and used to update records of the student table in the database.
- This method is also written in DAO classes related to entities of Lecturer Registration, Supervisor Registration, Faculty Registration, Program Registration, Course Registration, Committee Registration, Student - Course (Examination Details), Login Registration, Admin Registration.

```
public Student searchstudentbyid(int id)//Method used to used records of the table student
                                          which in the database Tithi
    {
      try
      {
        string sql = "select * from student where studentid = (" + id + ")";//sql query part
                        which is used to search records of the student table of database Tithi

        SqlCommand cmd = new SqlCommand(sql, m_con);//sql command used to connect
                                                      to the database
        m_con.Open();

        SqlDataReader dreader = cmd.ExecuteReader();


        if(dreader.Read())
        {
          std1/*Student object*/ = new Student((int)dreader[0], dreader[1].ToString(),
          dreader[2]. ToString(), dreader[3].ToString());//data in the student table of the
          database is read as an array
        }
        dreader.Close();

      }
      finally
      {
        m_con.Close();
      }
      return std1;//returning the student object
}
```

- Public method given above is written in the studentDAO class of the c sharp code to search data which is in the student table of database “Tithi”. This method returns a student object(std1).
- This method is also written in DAO classes related to entities of Lecturer Registration, Supervisor Registration, Faculty Registration, Program Registration, Course Registration, Committee Registration, Student - Course (Examination Details), Login Registration, Admin Registration.

```
private void button1_Click/*insert button used to add new record of a student*/ (object
sender, EventArgs e)
    {
      try
      {
        Student sob = new Student(int.Parse(textBox1.Text), (textBox2.Text),
                                  (textBox3.Text), (textBox4.Text)); //passing values in
                                  text boxes as a student object

        sdao.insertnewstudent(sob); //passing student object to the insert student method

        MessageBox.Show("Recorded Successfully", "Student Registration");

      }

      catch
      {
        MessageBox.Show ("Error", "Student Registration");
      }

    }
```

- The above code is written for the insert button of the Student Registration interface to pass the inserted values in the textbox as a student object to the database to record it in the table.
- This code is written in all other Insert Buttons of the forms related to entities of the Information Management System which includes, Lecturer Registration Interface, Supervisor Registration Interface, Faculty Registration Interface, Program Registration Interface, Course Registration Interface, Committee Registration Interface, Student – Lecturer Registration Interface, Lecturer - Course Registration Interface, Lecturer - Committee Registration Interface, Student - Course (Examination Details) Interface, Login Registration Interface, Admin Registration Interface.

```
private void button2_Click/*"delete button" - used to delete records from the student table in the
database*/ (object sender, EventArgs e)
    {
      try
      {
        sdao.deletestudentbyid(int.Parse(textBox1.Text)); //passing the value in textBox1 to the
                                                            delete method in the StudentDAO

        MessageBox.Show("Deleted Successfully", "Student Registration");
      }
      catch
      {

        MessageBox.Show("Error", "Student Registration");

      }
    }
```

- The above code is written in the delete button of the student interface to pass the value in textBox1 to the delete method in StudentDAO class.
- This code is written in all other Delete Buttons of the forms related to entities of the Information Management System which includes, Lecturer Registration Interface, Supervisor Registration Interface, Faculty Registration Interface, Program Registration Interface, Course Registration Interface, Committee Registration Interface, Student – Lecturer Registration Interface, Lecturer - Course Registration Interface, Lecturer - Committee Registration Interface, Student - Course (Examination Details) Interface, Login Registration Interface, Admin Registration Interface.

```
private void button3_Click/*Update Button" - used to update data from the student table*/ (object
sender, EventArgs e)
    {
      try
      {
        Student sob = new Student(int.Parse(textBox1.Text), (textBox2.Text),
                                  (textBox3.Text), (textBox4.Text)); //making student object
                                   from the inserted data

        sdao.updatestudent(sob);//passing student object to the update method in StudentDAO

        MessageBox.Show("Updated Successfully", "Student Registration");
      }
      catch
      {

        MessageBox.Show("Error", "Student Registration");
      }
```

- The above code is written in the Update button of the student interface and is used to pass the values inserted in the textboxes to the Update method of the StudentDAO class.
- This code is written in all other Update Buttons of the forms related to entities of the Information Management System which includes, Lecturer Registration Interface, Supervisor Registration Interface, Faculty Registration Interface, Program Registration Interface, Course Registration Interface, Committee Registration Interface, Student - Course (Examination Details) Interface, Login Registration Interface, Admin Registration Interface.

```
private void button4_Click/*"Search Buttton"- used to search records of the student table*/
(object sender, EventArgs e)
    {
      try
      {
        Student sob/*student object*/ =
        sdao.searchstudentbyid (int.Parse(textBox1.Text)/*Studentid*/);//passing value
        in   textBox1 to the searchstudentby id method

        textBox1.Text = sob.getid().ToString();
        textBox2.Text/*studentid*/ = sob.getname();
        textBox3.Text/*studentName*/ = sob.getdob();
        textBox4.Text/*date of birth*/ = sob.getyenrolled();

        MessageBox.Show("Search Successful", "Student Registration");

      }
      catch
      {

        MessageBox.Show ("Error", "Student Registration");

      }

    }
```

- The above is written in the Search button of the student interface which passes the value inserted in the textBox1 to the searchbyid method in the StudentDAO class to read data according to the id inserted and display the record of the student according to the ID in the Student Registration interface.
- This code is written in all other Search Buttons of the forms related to entities of the Information Management System which includes, Lecturer Registration Interface, Supervisor Registration Interface, Faculty Registration Interface, Program Registration Interface, Course Registration Interface, Committee Registration Interface, Student - Course (Examination Details) Interface, Login Registration Interface, Admin Registration Interface.

3.2)

Error Handling –

One way of Error handling is finding the errors and correct those errors during the application development so that they don't occur when running the code. But most of the time you may be unsure whether errors may occur or not. Therefore, in this situation which you cannot ensure that the errors will not occur, you can provide a solution code to deal with those errors in a way that it would not interrupt the usual execution of the code. [6]

```
private void button1_Click/*insert button used to add new record of a student*/ (object sender,
EventArgs e)
    {
      try
      {
        Student sob = new Student(int.Parse(textBox1.Text), (textBox2.Text),
                                  (textBox3.Text), (textBox4.Text)); //passing values in
                                  text boxes as a student object

        sdao.insertnewstudent(sob); //passing student object to the insert student method

        MessageBox.Show("Recorded Successfully", "Student Registration");

      }

      catch
      {
        MessageBox.Show ("Error", "Student Registration");
      }

    }
```

- Given above is an example for error handling which was implemented in the code written in the Insert button of Student registration Interface.

- In the above scenario if something is wrong with the entered data or with coding related to insertion of data, it will display an error message and if the insertion is successful, it will display a successful message as well.
- This method of error handling is used in all other the buttons of interfaces in the Information Management System of Stafford University.
- These Interfaces are namely,
    - Lecturer Registration Interface,
    - Supervisor Registration Interface,
    - Faculty Registration Interface,
    - Program Registration Interface,
    - Course Registration Interface,
    - Committee Registration Interface,
    - Student - Lecturer Registration Interface,
    - Lecturer - Course Registration Interface,
    - Lecturer - Committee Registration Interface,
    - Student - Course (Examination Details) Interface,
    - Login Registration Interface,
    - Admin Registration Interface.

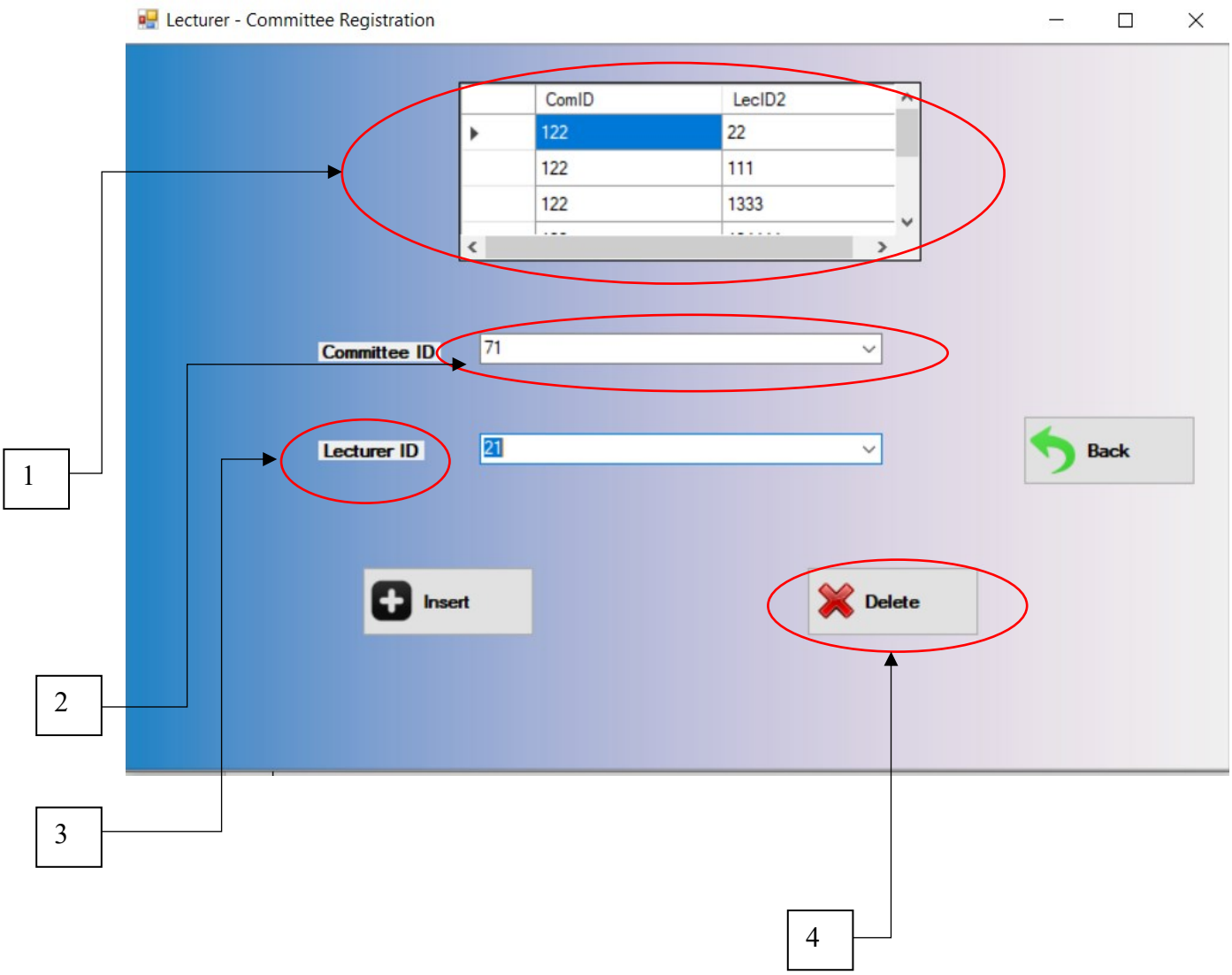
Advantages of Error Handling [7] –

- Separating error handling code from the main code.
- It indicates error by reporting the problems in the methods called.
- Errors are being categorized.

Disadvantages of Error Handling[8] –

- Can trap only runtime errors. Therefore, errors in SQL such as “Table does not exist” cannot be detected.
- It can mask the statements which caused the error.

3.3)



1 – Data Grid View

2 – Combo Box

3 – Label

4 – Button

Data Grid View – [13]

Data grid view is used to display the records available in the database table in the current situation to the user through a view shown on the form.

You can display a small amount of read only data as well as a large amount of editable data through this tool. This also allows the user to display data in master – details view.

Advantage – It's easy to handle the database table since it shows us what's there in the table.

Combo Box – [14]

Combo Box is used give user a selection option for a certain attribute in the table that has to enter data to using a dropdown list.

Properties of combo box are,

- Allow Selection,
- Auto Complete Custom source,
- Auto complete mode,
- Auto Complete Source,
- Data Bindings,
- Data Manager,
- Data Source,
- Dropdown height,
- Dropdown Style etc.

Methods of Combo box Control are,

- Dropdown,
- Dropdown Closed,
- Dropdown Style Changed etc.

Label – [15]

Label is used to display text on the form usually to give a meaning to combo box, Text Box etc.

It will not change its content during the runtime.

Properties of Label are,

- Auto Size,
- Border Style,
- Flat Style,
- Font,
- Font Height,
- Fore Colour,

- Preferred Height etc.

Button – [16]

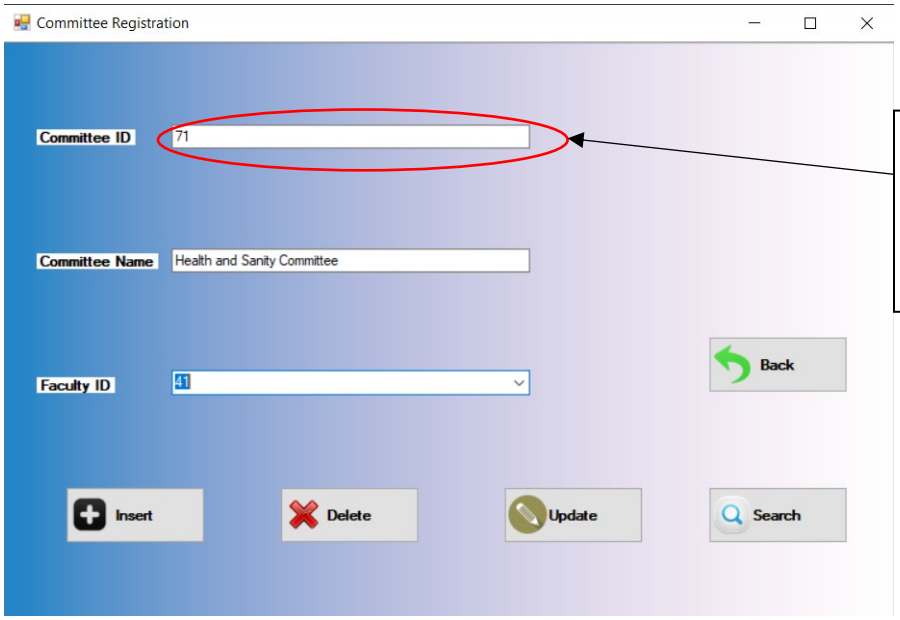
Button is used to allow user to access a certain function of the system by just clicking it.

Properties of Button are,

- Auto Size Mode,
- Back Color,
- Background image,
- Dialog Result,
- Fore Color,
- Image etc.

Methods of Button Control are,

- Get preferred size
- Notify default,
- Select etc.



Text Box – Used to give space for the user to enter text, usually to manipulate data in the database tables.

Task 4

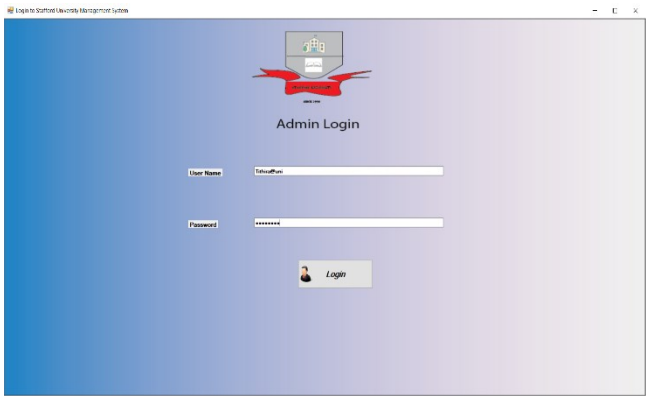
4.1) Test Plan to Support the Documentation of Testing Process.

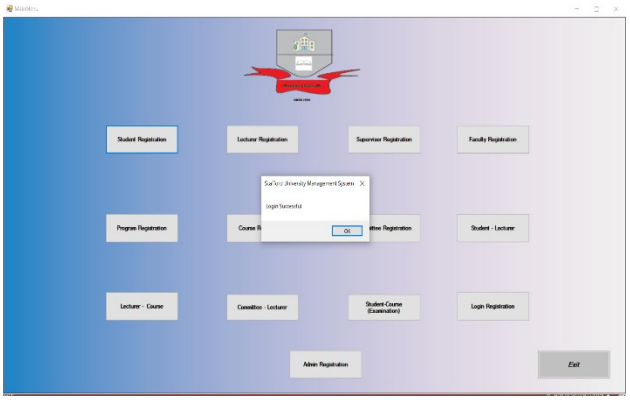
Information Management System of Stafford University	
Test Plan ID	01
Brief Introduction of the Information Management System.	The Information Management System of the Stafford University is mainly designed to manage all the information related to the university.
Introduction to the Testing Objectives.	This document mainly describes the plan for testing the Information Management System of Stafford University and this Document supports the following objectives as well, • Testing the features of the System. • Recommended Test Requirements. • Describing the recommended test strategies. • List the deliverable elements of the test activities.
Testing items	Main Functions of the System.
Features to be tested.	Admin Login to the System, Inserting Records, Deleting Records, Updating records, Searching Records.
Testing Environment	PC, Microsoft SQL Server Management Studio 2019, Visual Studio 2019.
Testing Approach	Black Box Testing.
Testing Tasks	Test Planning, Test Design, Test Development, Test Execution, Test Evaluation.
Test deliverables	Test Plan, Test Environment, Test Result (Test Cases)
Schedule	9th September 2020, 7.57pm to 9.55pm

4.2)

Project Name – Information Management System of Stafford university	
Test Case	
Test Unit – Admin Login	Tester – Tithira Withanaarachchi
Test Case ID - 01	Test Type – Black Box Testing
Description – Connecting and logging to the Information Management System of Stafford University.	Test Executed by – Tithira Withanaarachchi
	Test Execution Date – 11th September 2020
Title – Logging to the Information Management System.	Test Execution Time – 7.57pm to 8.13pm

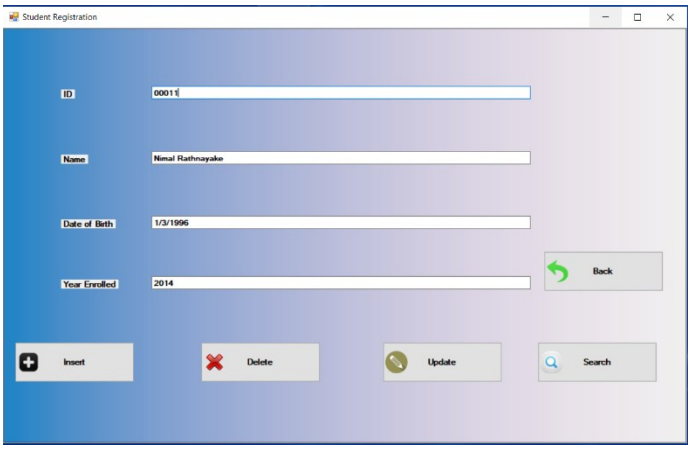
Step No	Test Steps	Testing Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
01	Enter Valid Credentials					
02	Perform Validation	UserName: Tithira@uni Password: tithi123	Entering the mainmenu successfully and getting a login successful message	Login successful	Pass	

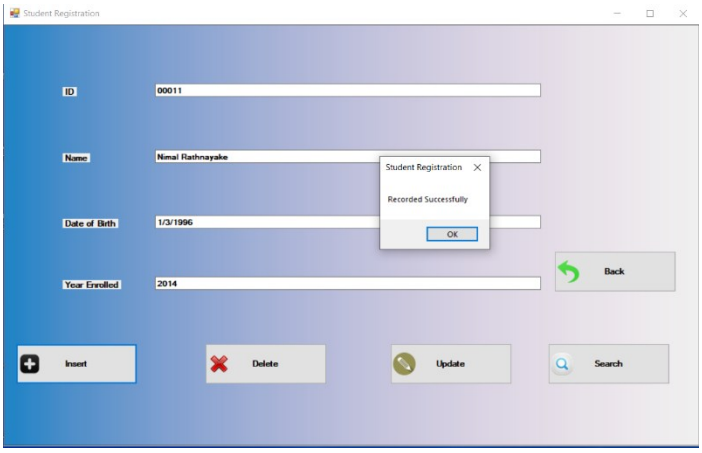




Project Name – Information Management System of Stafford University	
Test Case	
Test Unit – Inserting Records	Tester – Tithira Withanaarachchi
Test Case ID - 02	Test Type – Black Box Testing
Description – Insert a new record to the table in database.	Test Executed by – Tithira Withanaarachchi
	Test Execution Date – 11th September 2020
Title –Inserting Record .	Test Execution Time – 8.17pm to 8.25pm

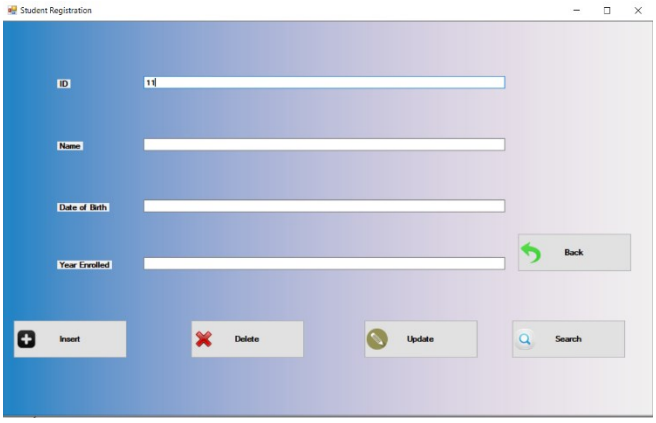
Step No	Test Steps	Testing Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
01	Login to the System and access related form to the table.					
02	Insert records	Checking the Insert Button Click	Inserting a new record to the table in the database.	Insert Successful	Pass	

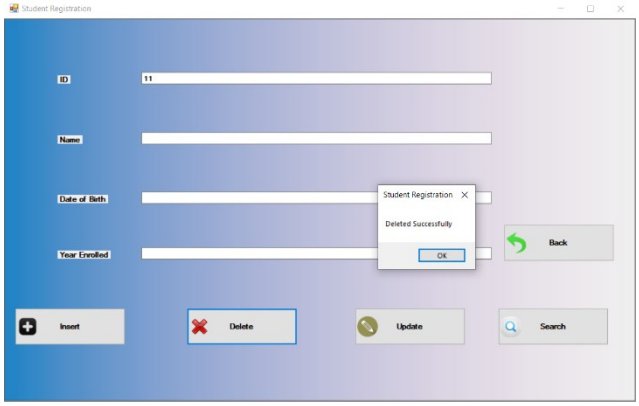




Project Name – Information Management System of Stafford university	
Test Case	
Test Unit – Deleting Records	Tester – Tithira Withanaarachchi
Test Case ID - 03	Test Type – Black Box Testing
Description – Deleting a particular record according to the ID given by the user.	Test Executed by – Tithira Withanaarachchi
	Test Execution Date – 11th September 2020
Title –Delete by ID .	Test Execution Time – 9.08.pm to 9.15pm

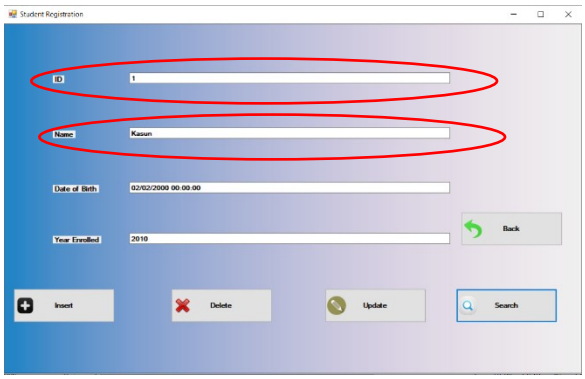
Step No	Test Steps	Testing Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
01	Login to the System and access related form to the table.					
02	Delete Record	Checking the Delete Button Click	Deleting a record of a table in the database.	Delete Successful	Pass	

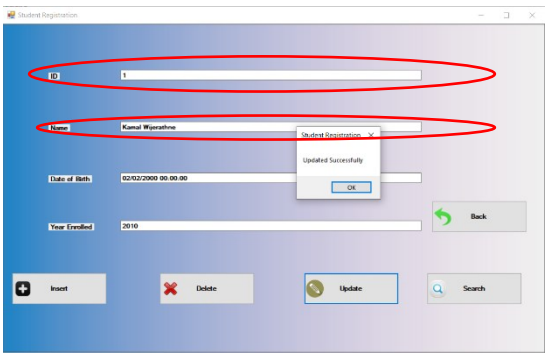




Project Name – Information Management System of Stafford university	
Test Case	
Test Unit – Updating Records	Tester – Tithira Withanaarachchi
Test Case ID - 04	Test Type – Black Box Testing
Description – update a record in the table in database.	Test Executed by – Tithira Withanaarachchi
	Test Execution Date – 11th September 2020
Title –Update Record .	Test Execution Time – 9.42pm to 9.50pm

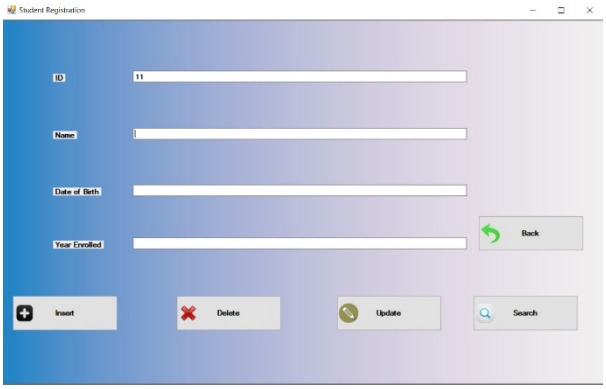
Step No	Test Steps	Testing Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
01	Login to the System and access related form to the table.					
02	Update Record	Checking the Update Button Click	Updating a record of a table in the database.	Update Successful	Pass	

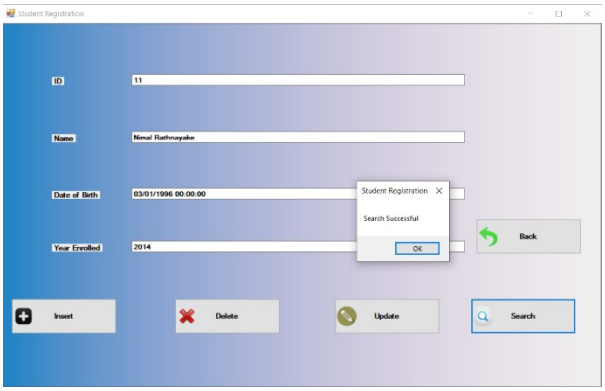




Project Name – Information Management System of Stafford university	
Test Case	
Test Unit – Searching Records	Tester – Tithira Withanaarachchi
Test Case ID - 05	Test Type – Black Box Testing
Description – Search a particular record according to the ID given	Test Executed by – Tithira Withanaarachchi
	Test Execution Date – 11th September 2020
Title –Search by ID	Test Execution Time – 9.50 pm to 9.55pm

Step No	Test Steps	Testing Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
01	Login to the System and access related form to the table.					
02	Search Record	Checking the Search Button Click	Display the searched record by ID.	Record Displayed	Pass	





4.3)

User Documentation

Documentation indicates how the product or software operates to the user. This document assists the user to operate the product. This document can be considered as a part of the product as well.

User Documentation is important because it provides,

- How to use the software.
- Features of your Software.
- Tips and tricks of your software.
- How to resolve common problems of your software.[9], [10], [11]

User's Guide for Stafford University Information Management System.

- **General Information**

1.1 **System Overview**

Information Management System of Stafford University

- A software System Based on the Windows Platform.
- Graphically created interfaces for system notifications.
- Store data of Students, Lecturers, Admins, Supervisors, Faculties, Programs, Courses, Committees etc.
- System Name: Information Management System of Stafford University.
- System Code: C# and SQL Server.
- System Category:
  Major Application: Performs clearly defined functions for storing information about all the entities of the Stafford University.

Authorized Use Permission

Stafford University provides you with access to a variety of resources on this system including,

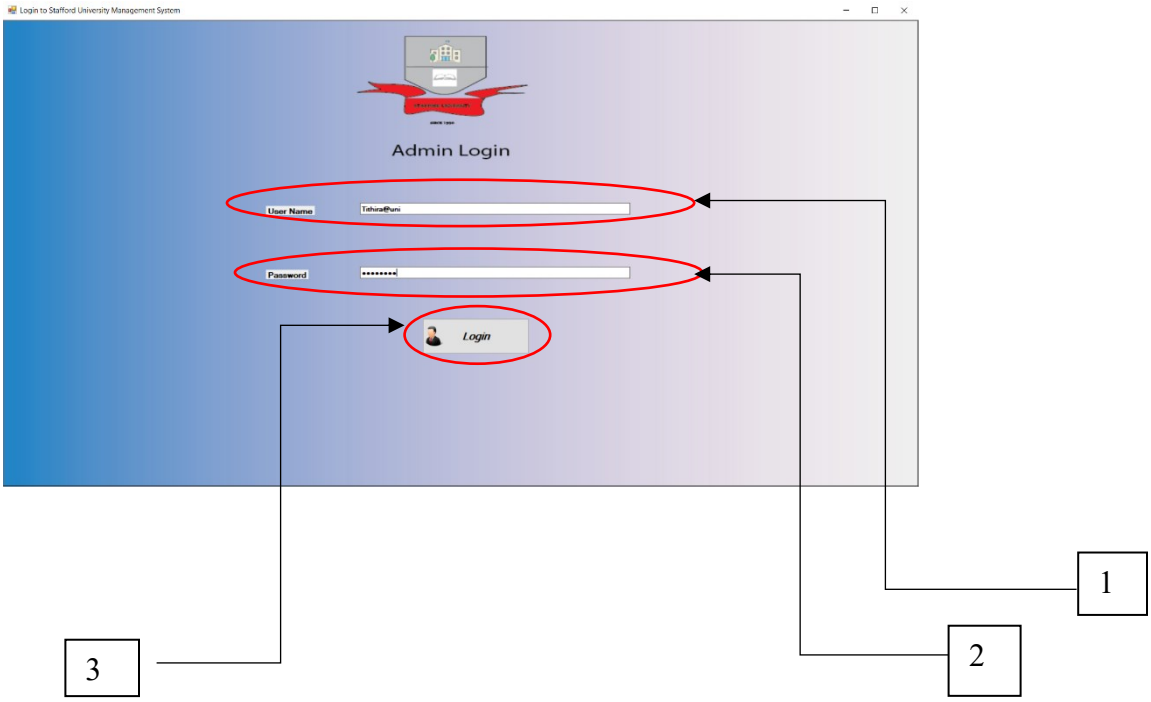
- student registration,
- Lecturer Registration,
- Supervisor Registration,
- Faculty Registration,
- Program Registration,
- Course Registration,
- Committee Registration,
- Student - Lecturer Registration,
- Lecturer - Course Registration,
- Lecturer - Committee Registration,
- Student - Course (Examination Details),
- Login Registration,
- Admin Registration.

2. Getting Started

This section provides a general walkthrough of the system from initiation through exit. The logical arrangement of the information will enable the users to understand the sequence and flow of the system. Have used screen prints to show the examples of text under each heading.

2.1 Admin Login

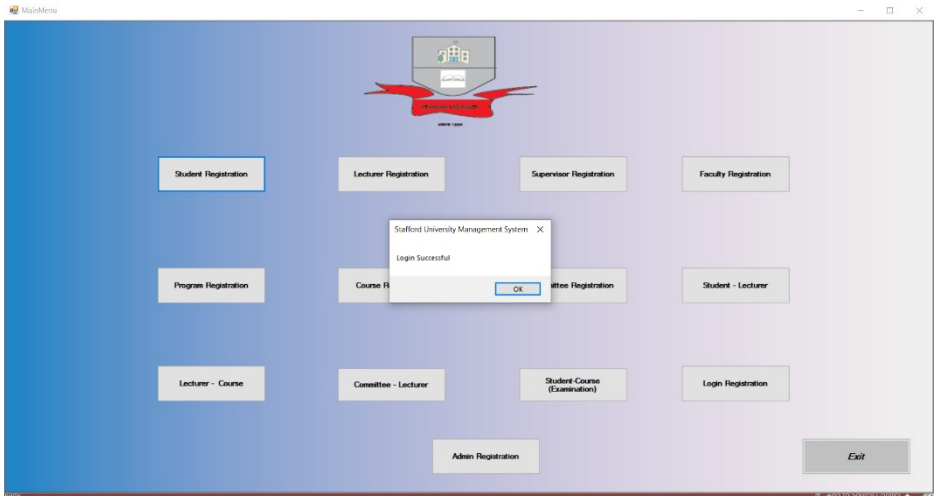
A user ID and the password is needed to log in to the Information Management System.



1 – Enter User Name.

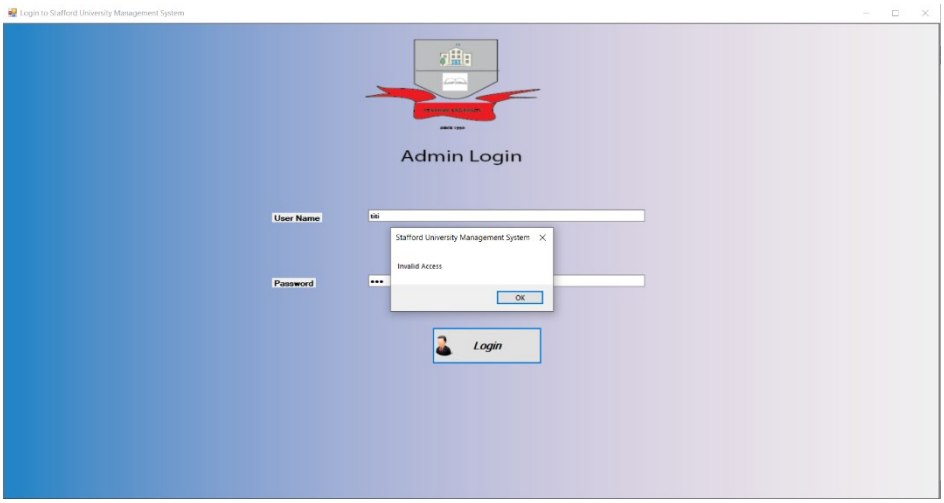
2 – Enter Password.

3 – Click the login Button.



If all the credentials are correct then the system will display the Main Menu giving a popup screen saying “login successful”. Click OK.

If there is something wrong with the credentials entered it will show a popup screen saying that “Invalid Access”.



2.2 System Menu

When Login to the system successfully, it will show up the Main menu which includes all the buttons which gives access to all the forms which allows to manipulate data related to each entity.

Click the entity button you wish to manipulate data in. Therefore, the form related to manipulating data related to the entity that you wish to manipulate data is being showed up.

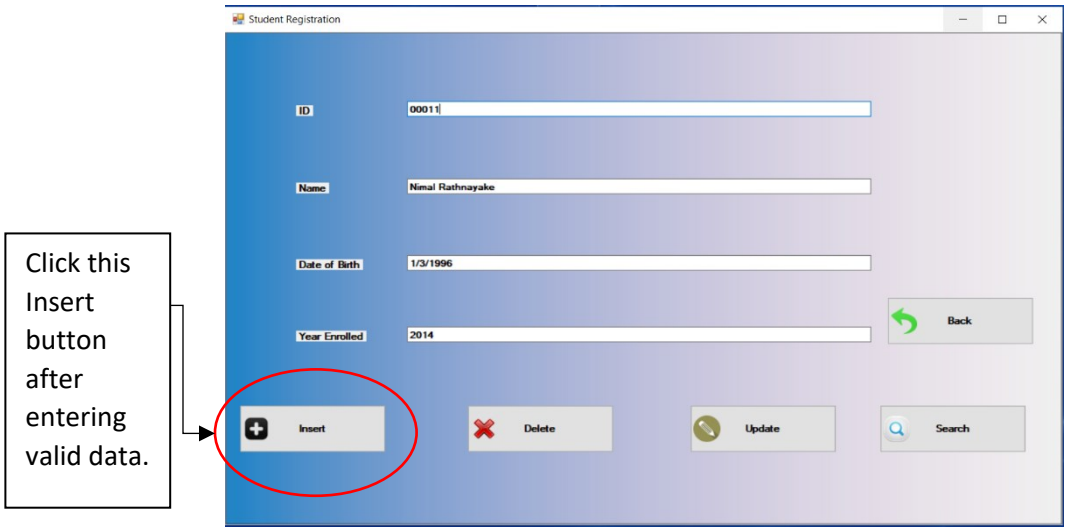
1 - For the illustration purpose, I have selected the "Student Registration" button for data manipulating in the student registration table of database.

You can click any button from the Main Menu to manipulate data as you wish related to any entity.

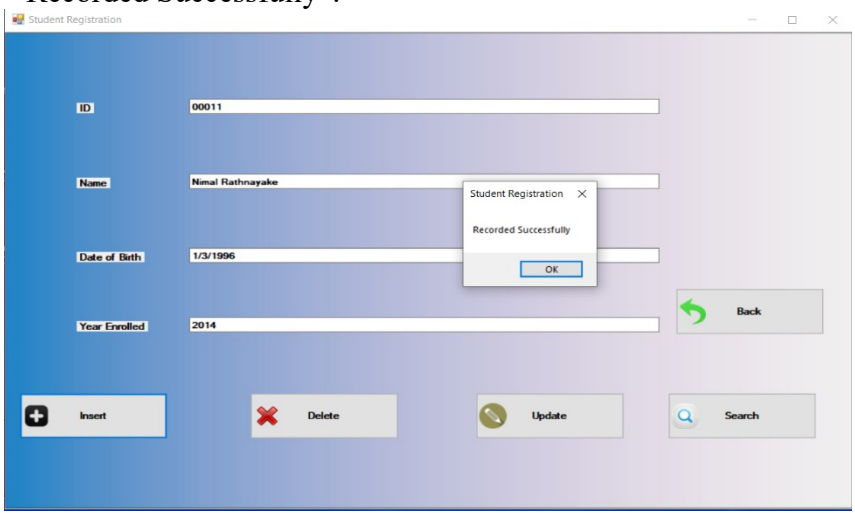
2.3 Data Manipulating Form

a. Form with sample Data values for inserting a new record

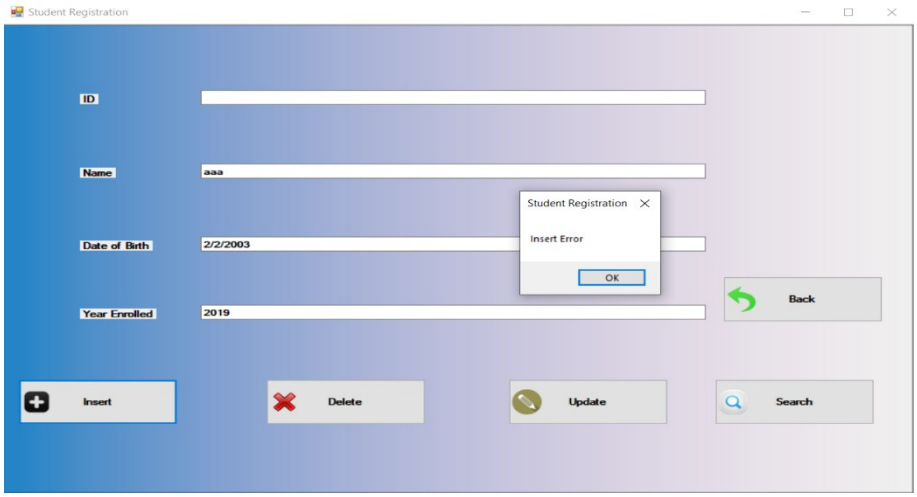
i. Data which have to insert to the table should be typed in the relevant space for each attribute.
ID = integer type data which is the identification number of a particular record.
Name = Name of the student which contains string type data.
Date of Birth = Date of birth of the student which contains string type data and should be in the date structure.
Year Enrolled = Year that the student has enrolled to the university which contains string type data.



ii. Popup Message is displayed when recorded the data successfully, saying that "Recorded Successfully".

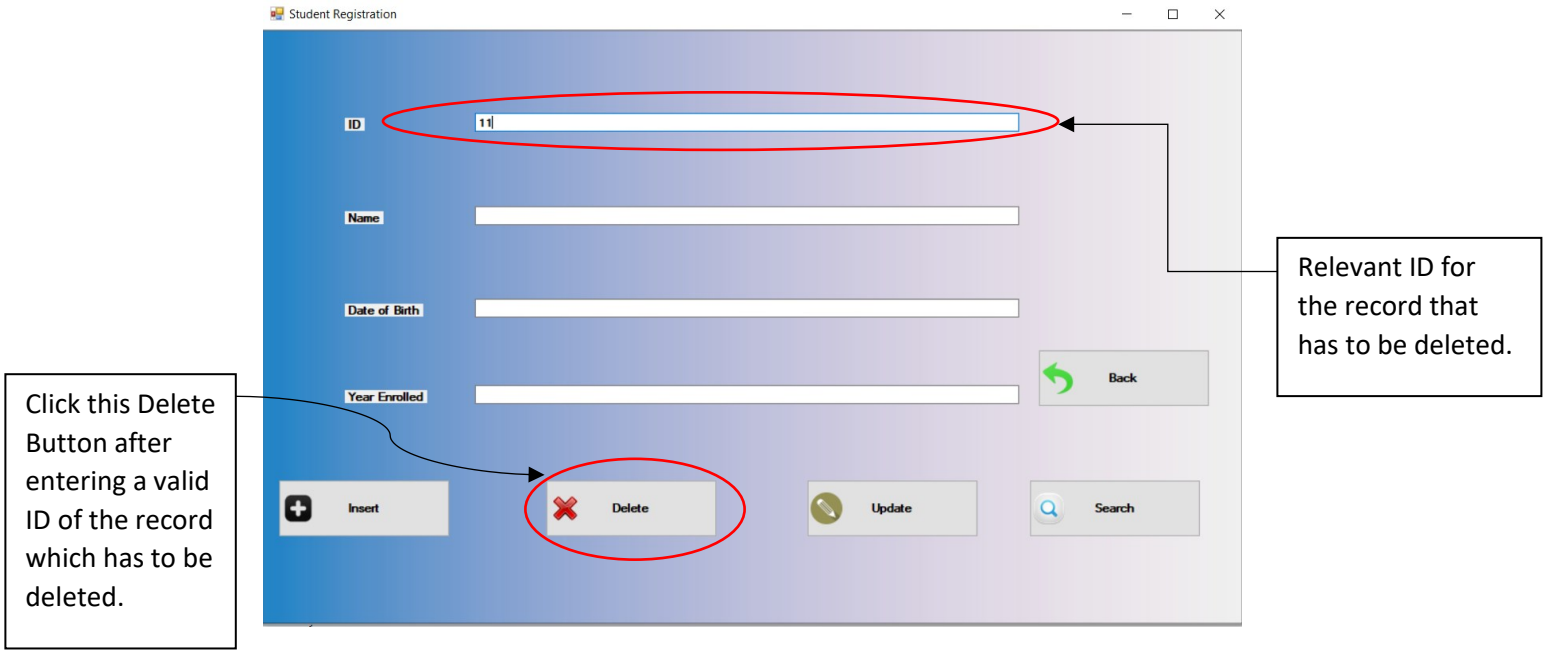


iii. If there is something wrong with the entered data, it will show a popup message saying “Insert Error”.

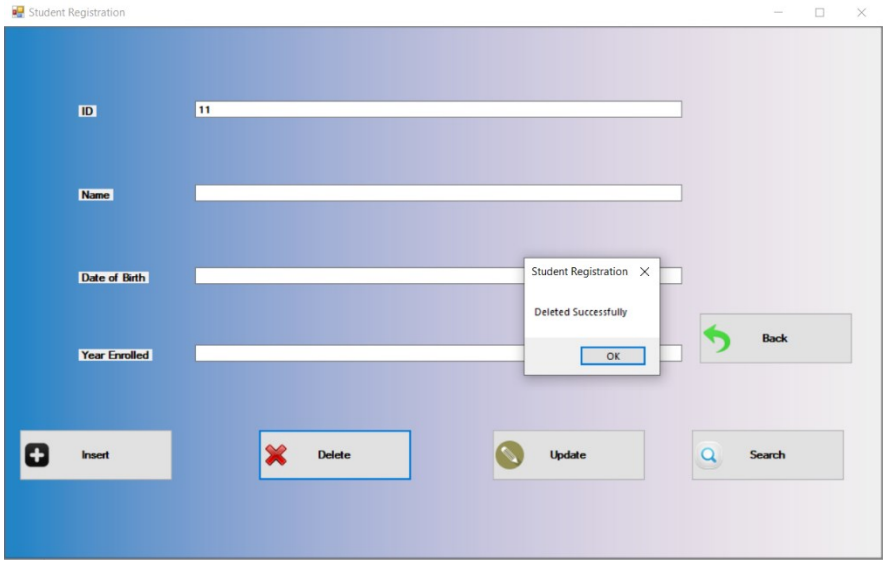


b. **Form with Sample Records for Deleting a Record**

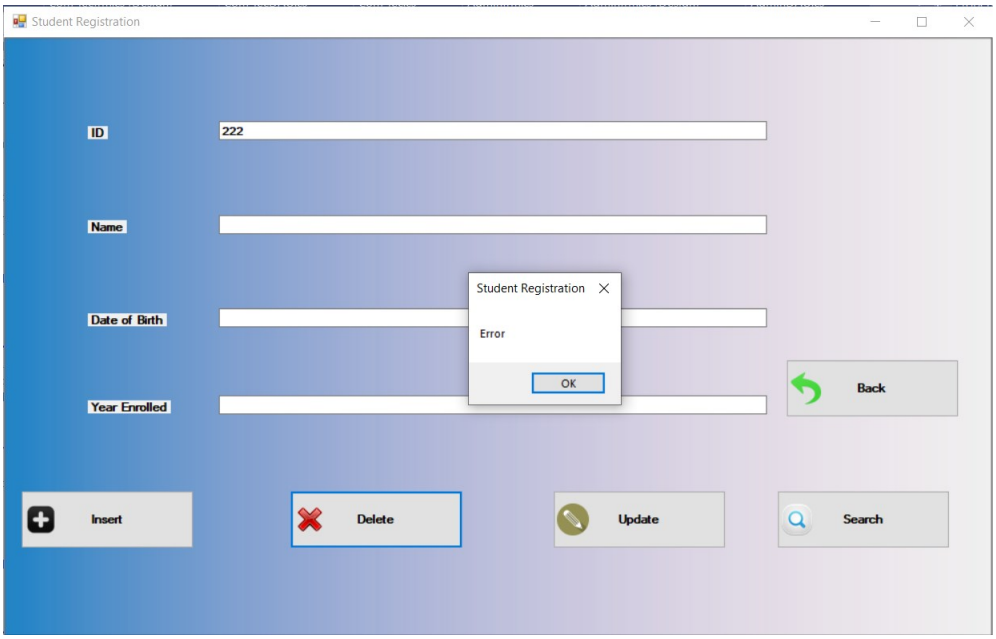
i. To delete a record, the user should enter the relevant ID for the record which has to be deleted and then Click the Delete Button



ii. If the entered ID is correct (if the ID is an existing value in the table) it will display a popup message saying “Deleted Successfully”.

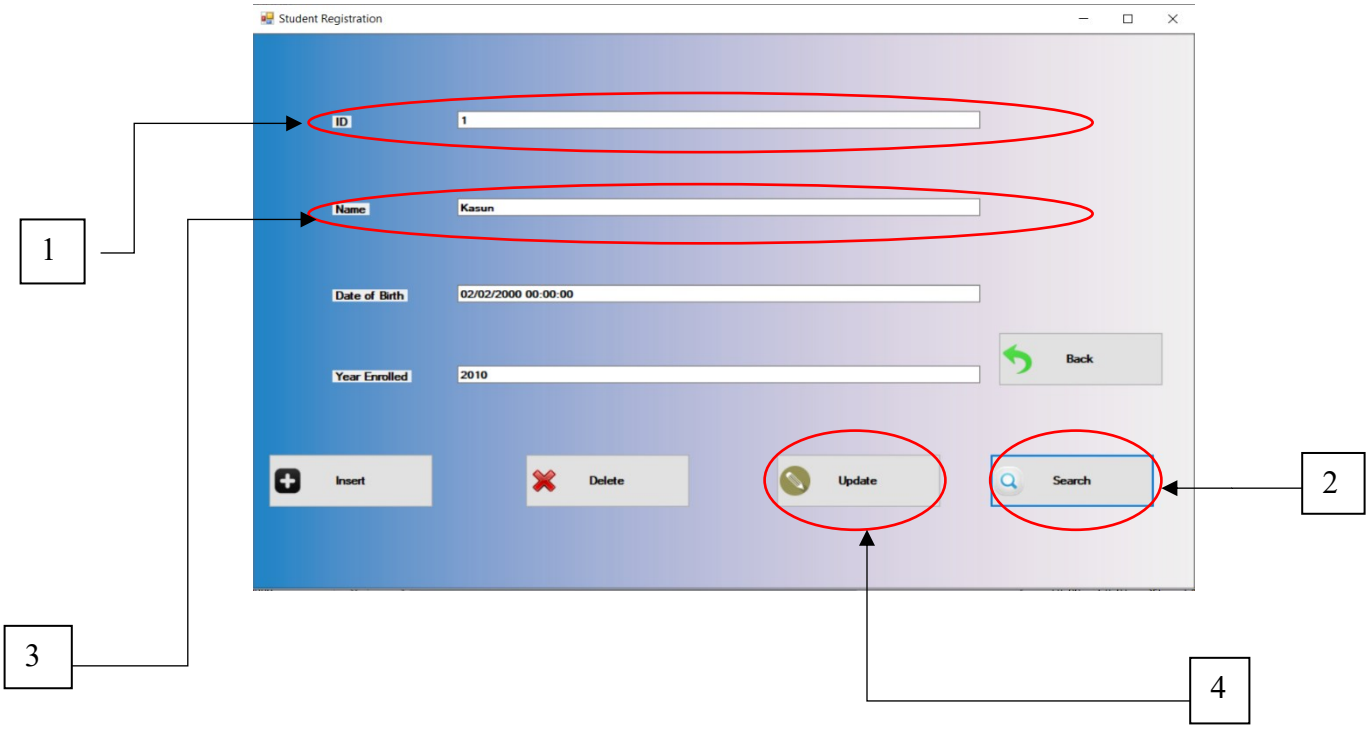


iii. If something wrong with the entered ID (or if the ID is not available) it will show a popup message saying “Error”.



c. **Form with Sample Records to Update Data in the Table**

i. Before Updating the user has to search the record through its ID, there after the user can manipulate those data in that particular record while considering the data type restrictions and other restriction stated for the insertion function. Finally Click the Update Button.



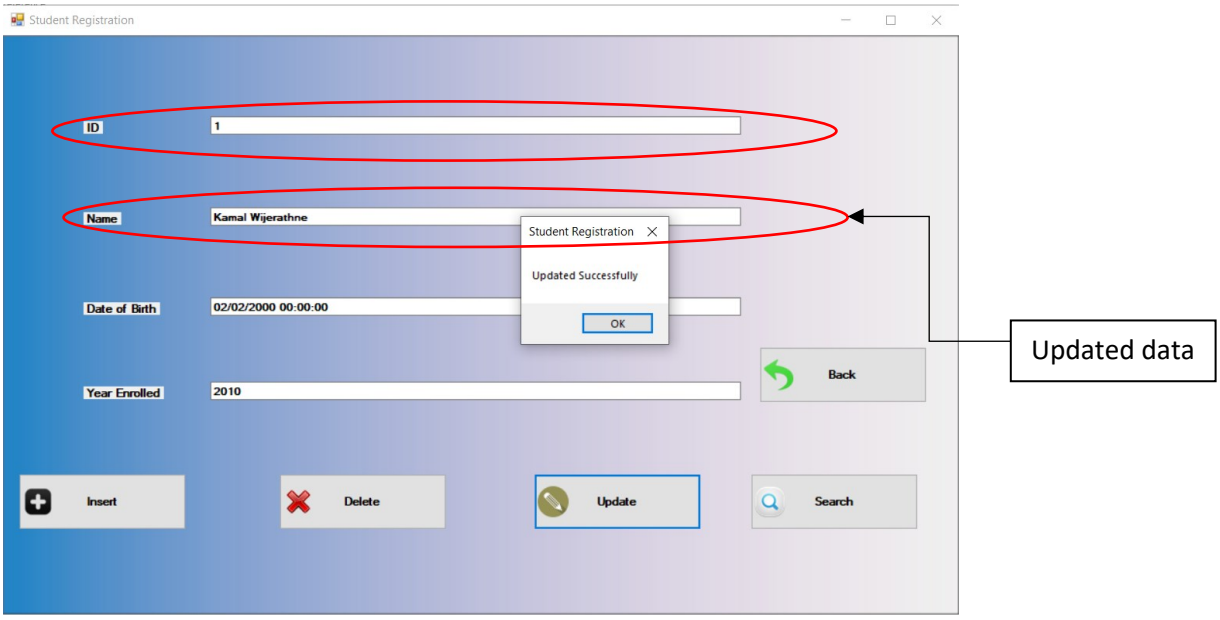
1 – First Enter the relevant ID related to the record that is needed to be manipulated.

2 – Then search it through the search button.

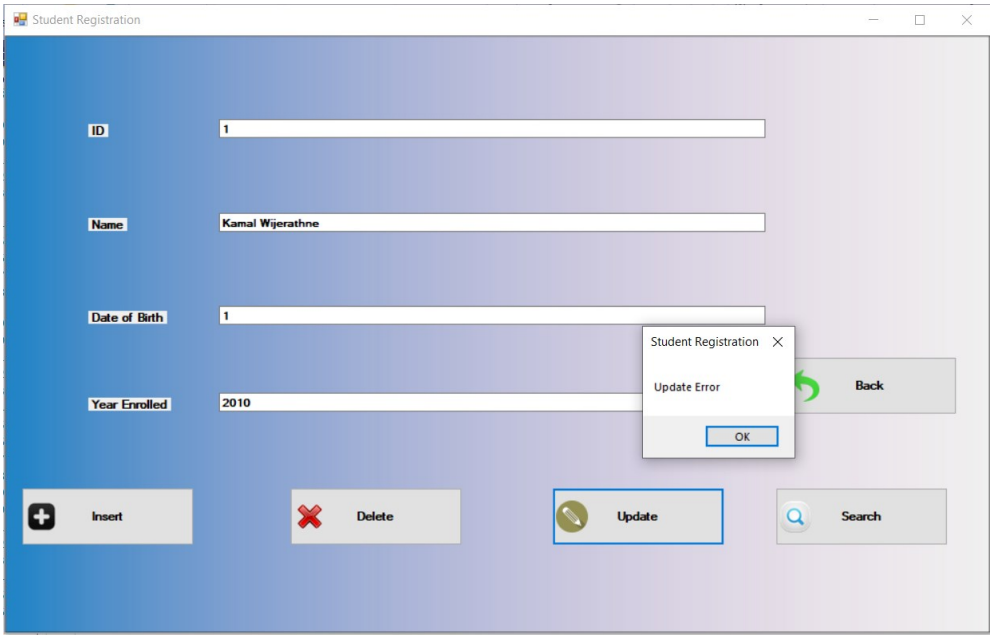
3 - Then change the data that you have to manipulate except the ID of the record. (since you are not allowed to change the identifier of the record.)

4 – Finally Click the Update button.

ii. If the manipulated data are according to the restrictions, it will display a popup message saying “Updated Successfully”

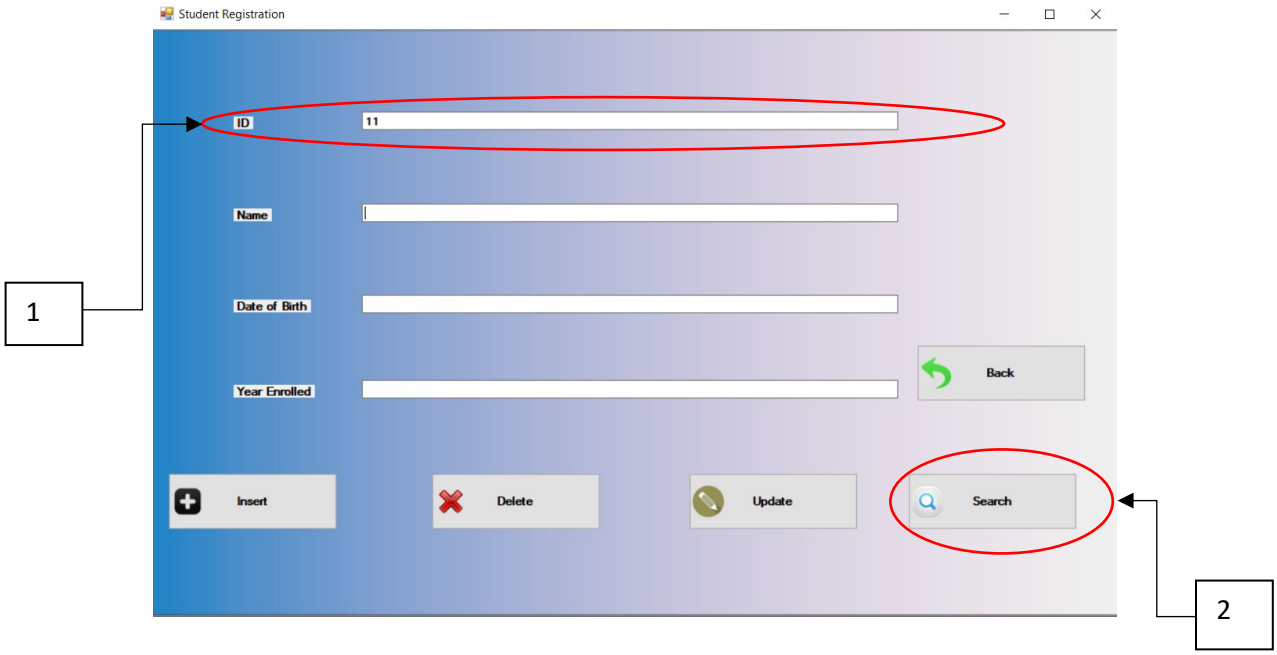


iii. If there is something wrong with the manipulated data (Updated data) it will show a popup message indicating an error.



d. Form with Sample Records to Search Data

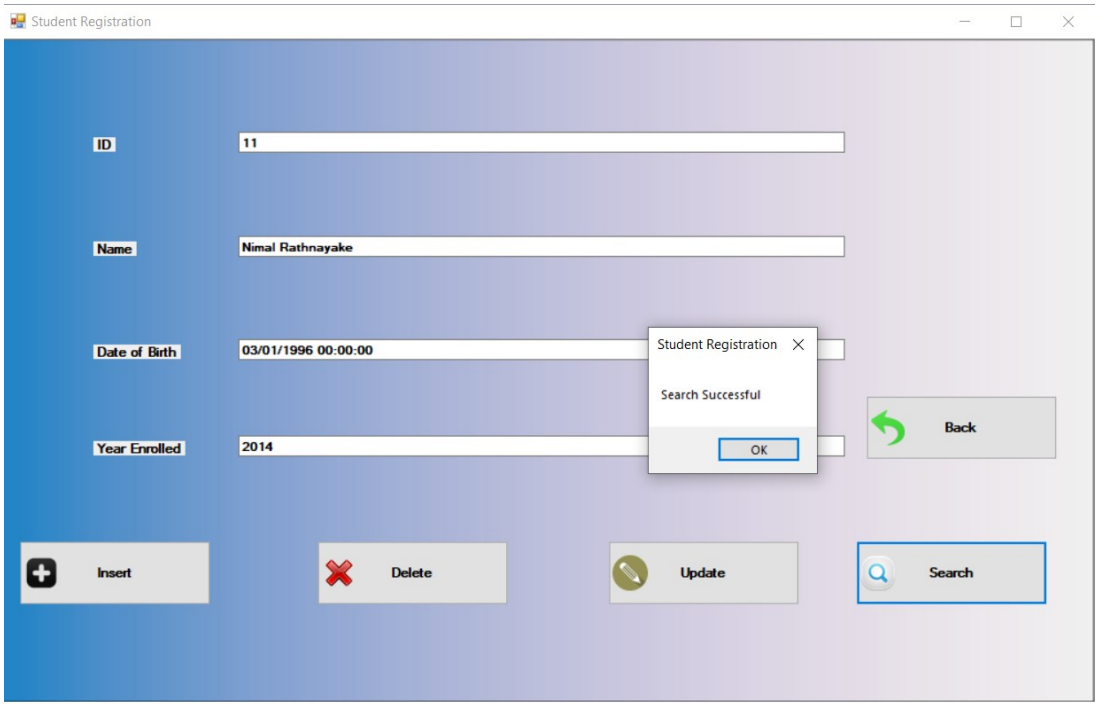
i. Type the ID related to the record that has to be searched, in the space given(textbox1) (ID)



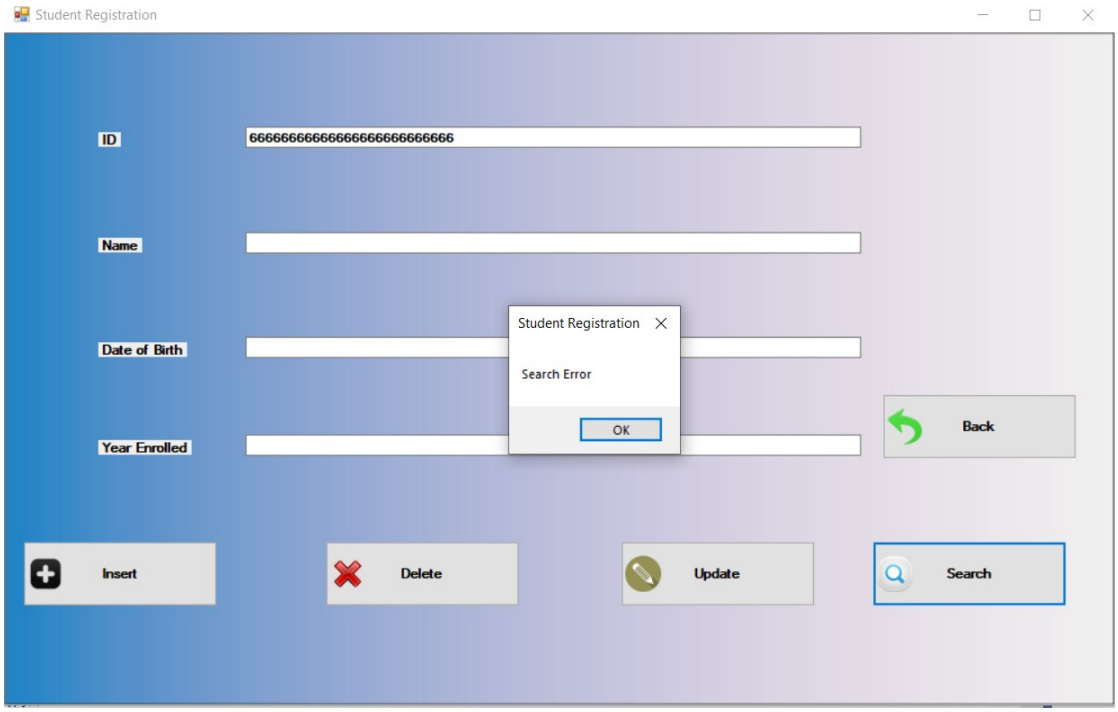
1 – Insert the ID

2 – Click the search button.

ii. If the ID is an available one in the table it will display the relevant record in the text Boxes and display a popup message saying “Search Successful”



iii. If the entered ID is not available in the table it will display an error through a popup message.



References

[1]"Redirect Notice", *Google.com*, 2020. [Online]. Available: https://www.google.com/url?sa=i&url=http%3A%2F%2Fwww.oogazone.com%2F2019%2Funique-open-book-icon-vector-drawing%2F&psig=AOvVaw3xCH6FeCFdcQQL-R6iSb-F&ust=1599638019198000&source=images&cd=vfe&ved=0CAIQjRxqFwoTCNDs4KKK2esCFQAAAAAdAAAAABAK. [Accessed: 07- Sep- 2020].

[2]"5,034 Free vector icons of university", *Flaticon*, 2020. [Online]. Available: https://www.flaticon.com/free-icons/university. [Accessed: 07- Sep- 2020].

[3]"Login Form With SQL in C#", *C-sharpcorner.com*, 2020. [Online]. Available: https://www.c-sharpcorner.com/UploadFile/9582c9/login-form-with-sql-in-C-Sharp/. [Accessed: 10- Sep- 2020].

[4]"User 16x16 Icons - Download 2157 Free User 16x16 icons here", *Iconarchive.com*, 2020. [Online]. Available: https://www.iconarchive.com/tag/user-16x16. [Accessed: 14- Sep- 2020].

[5]"Login Form With SQL in C#", *C-sharpcorner.com*, 2020. [Online]. Available: https://www.c-sharpcorner.com/UploadFile/9582c9/login-form-with-sql-in-C-Sharp/. [Accessed: 14- Sep- 2020].

[6]B. Perkins, J. Hammer and J. Reid, *Beginning C# 7 programming with Visual Studio 2017*. .

[7]"What happens when exception is thrown by main method?", *R4r.in*, 2020. [Online]. Available: http://www.r4r.in/corejava/interview-question-answers/?request_id=1-11&question_id=252. [Accessed: 14- Sep- 2020]

[8]"Error Handling", *Docs.oracle.com*, 2020. [Online]. Available: https://docs.oracle.com/cd/A57673_01/DOC/server/doc/PLS23/ch6.htm#:~:text=will%20be%20handled.-,Disadvantages,or%20view%20does%20not%20exist. [Accessed: 14- Sep- 2020].

[9]"Product documentation: how to ensure it is high quality? Examples | 3di", *3di Information Solutions: technical communication, translation and localization*, 2020. [Online]. Available: https://3di-info.com/how-to-ensure-your-product-documentation-is-high-quality/. [Accessed: 14- Sep- 2020].

[11]"User documentation - Computer Science Wiki", *Computersciencewiki.org*, 2020. [Online]. Available: https://computersciencewiki.org/index.php/User_documentation. [Accessed: 14- Sep- 2020].

[12]"Components of .Net Framework, CLR, CTS, CLS, Base Class Library", *Developerin.net*, 2020. [Online]. Available: http://www.developerin.net/a/39-Intro-to-.Net-FrameWork/23-Components-of-.Net-Framework. [Accessed: 14- Sep- 2020].

[13]"DataGridView Control In C#", *C-sharpcorner.com*, 2020. [Online]. Available: https://www.c-sharpcorner.com/uploadfile/raj1979/datagridview-control-in-C-Sharp/. [Accessed: 14- Sep- 2020].

[14]"VB.Net - ComboBox Control - Tutorialspoint", *Tutorialspoint.com*, 2020. [Online]. Available: https://www.tutorialspoint.com/vb.net/vb.net_combobox.htm. [Accessed: 14- Sep- 2020].

[15]"VB.Net - Label Control - Tutorialspoint", *Tutorialspoint.com*, 2020. [Online]. Available: https://www.tutorialspoint.com/vb.net/vb.net_label.htm. [Accessed: 14- Sep- 2020].

[16]"VB.Net - Button Control - Tutorialspoint", *Tutorialspoint.com*, 2020. [Online]. Available: https://www.tutorialspoint.com/vb.net/vb.net_button.htm. [Accessed: 14- Sep- 2020].